

A Direct-Sum Lower Bound for Private Learning of Disjoint Threshold Minors

July 27, 2026

1 Theoretical Setup

1.1 Problem setting

For an integer $r \geq 1$, write $[r] = \{1, \dots, r\}$. The logarithm $\log$ is natural, while $\log^*$ is the base-two iterated logarithm; changing either base changes only universal constants. For $t \in [N]$, define the lower-oriented threshold

$$\tau_t(q) = \mathbf{1}\{q \leq t\}, \quad q \in [N].$$

Let X be an arbitrary domain and let $C \subseteq \{0, 1\}^X$ be finite. A (k, N) disjoint-threshold minor consists of injections $\phi_j : [N] \rightarrow X$, $j \in [k]$, having pairwise disjoint images, and concepts $c_{\mathbf{t}} \in C$, indexed by $\mathbf{t} = (t_1, \dots, t_k) \in [N]^k$, such that

$$c_{\mathbf{t}}(\phi_j(q)) = \tau_{t_j}(q) \quad \text{for every } j \in [k] \text{ and } q \in [N].$$

For a labeled distribution Q on $Z_X := X \times \{0, 1\}$ and a hypothesis $h : X \rightarrow \{0, 1\}$, its zero-one risk is

$$L_Q(h) = \Pr_{(x,y) \sim Q} \{h(x) \neq y\}.$$

Two ordered samples of the same size are replacement adjacent when they differ in at most one record. The hypothesis space is equipped with a sigma-algebra for which every coordinate evaluation is measurable. A randomized kernel $A : Z_X^m \rightsquigarrow \{0, 1\}^X$ is (ε, δ) -differentially private if, for every measurable event E and every replacement-adjacent $S, S' \in Z_X^m$,

$$A(S)(E) \leq e^\varepsilon A(S')(E) + \delta, \quad A(S')(E) \leq e^\varepsilon A(S)(E) + \delta.$$

No properness, output-representation, or computational restriction is imposed.

For $c \in C$ and a distribution P on X , let $Q_{P,c}$ be the law of $(x, c(x))$ for $x \sim P$. The learner receives exactly m iid labeled examples; all probabilities below include both sample and learner randomness.

1.2 Assumptions

Assumption 1 (Disjoint-threshold restriction table). *Integers $k \geq 1$ and $N \geq 2$ and the finite class C admit injections $\phi_1, \dots, \phi_k$ and concepts $c_{\mathbf{t}}$ forming a (k, N) disjoint-threshold minor as defined above. This is a static restriction-table condition.*

Assumption 2 (Unrestricted private PAC antecedent). *For an integer $m \geq 1$, the kernel $A : Z_X^m \rightsquigarrow \{0, 1\}^X$ is replacement-adjacency $(\varepsilon_0, \delta_m)$ -differentially private and is distribution-free realizable (α_0, β_0) -PAC: for every $c \in C$ and every distribution P on X ,*

$$\Pr_{S \sim Q_{P,c}^m, h \sim A(S)} \{L_{Q_{P,c}}(h) > \alpha_0\} \leq \beta_0.$$

The output may be improper, and the learner may be computationally unbounded.

Assumption 3 (Fixed PAC and approximate-privacy scale). *The accuracy, confidence, and privacy parameters satisfy*

$$\alpha_0 = \frac{1}{128}, \quad \beta_0 = \frac{1}{32}, \quad 0 < \varepsilon_0 \leq 1, \quad 0 \leq \delta_m \leq \frac{c_\delta}{m^2 \log(em)},$$

where $c_\delta > 0$ is a scalar parameter. Thus this assumption is parameterized by c_δ ; every theorem citing it specifies the value being used.

1.3 Formalized goal

The objective is a fixed-sample lower bound in zero-one risk with a high-probability, distribution-free realizable-PAC antecedent. We prove that universal constants $c, c_\delta > 0$ and $N_0 \geq 2$ can be chosen so that every learner satisfying Assumptions 1–2 and Assumption 3 instantiated with that same c_δ , for every $k \geq 1$ and $N \geq N_0$, must satisfy $m \geq ck \log^* N$. The constants are independent of $C, X, k, N, m, \varepsilon_0, \delta_m$, the learner, and the minor embeddings.

2 Preliminaries

For $x > 1$, define $\log_2^{(0)} x = x$ and $\log_2^{(r+1)} x = \log_2(\log_2^{(r)} x)$ whenever the latter is defined. We use

$$\log^* x := \min\{r \geq 0 : \log_2^{(r)} x \leq 1\}.$$

This is the only asymptotic scale in the main theorem. The lower bound is deterministic once the fixed-sample PAC and privacy antecedents hold; the probability in the antecedent is the joint probability over the iid sample and the learner’s internal randomness, and the relevant metric is zero-one risk L_Q .

3 Main Theorem

Theorem 3.1 (Private direct-sum threshold-minor lower bound). *There exist universal constants $c, c_\delta > 0$ and an integer $N_0 \geq 2$ such that the following holds. For every $k \geq 1$, every $N \geq N_0$, every domain X , every finite $C \subseteq \{0, 1\}^X$, and every fixed-sample learner A satisfying Assumptions 1–2 and Assumption 3 instantiated with the same c_δ quantified in this theorem,*

$$m \geq ck \log^* N.$$

The conclusion is deterministic; the antecedent is high-probability, distribution-free realizable PAC in zero-one risk at the fixed sample size m . All displayed constants are independent of C, X, k, N, m , of ε_0, δ_m , and of the learner and minor table.

Corollary 3.1 (Exact one-chain specialization). *Under the assumptions of Theorem 3.1, if $k = 1$, then $m \geq c \log^* N$. In the reduction proving this statement the auxiliary sample size, sole-arm occupancy, and overflow event are defined by*

$$n := 2m + 12, \quad R := m, \quad \mathcal{O} := \{R > n\}.$$

Consequently $R = m < n$ and $\mathcal{O}$ is empty, there are no nonhidden-arm simulations, and the restricted threshold kernel inherits exactly $(\varepsilon_0, \delta_m)$ -privacy and the one-chain risk interface.

4 Proof Sketch

The proof begins with a finite one-dimensional threshold game. An arbitrary hypothesis is restricted to its predictions on $[N]$, which preserves both privacy and every risk in the game. Sampling n rows with replacement from an empirical database of size $9n$ gives an empirical wrapper whose expected empirical loss is exactly the finite-game risk. The corrected secrecy-of-the-sample calculation of [Bun et al. \(2015\)](#) preserves the original privacy parameters uniformly through $\varepsilon = 1$.

The homogeneous-set argument of [Alon et al. \(2018\)](#) is then rederived at expected loss $1/20$. A finite coloring records all prediction marginals, the finite Ramsey theorem of [Erdős and Rado \(1952\)](#) gives a large homogeneous set, and endpoint privacy comparisons yield a positive adjacent rise. Moving one record across the remaining ordered block produces a family of pairwise private output laws with an off-diagonal margin. A finite product experiment and binary-search tree turn this margin into an upper bound on the homogeneous set. Comparing the Ramsey lower bound and binary upper bound and inverting the resulting iterated logarithms gives universal constants $a_{\text{th}}, a_\delta > 0$ and N_{th} for the one-arm obstruction.

The finite set of private kernels is a compact convex polytope and each experiment risk is affine. Finite minimax therefore converts the algorithm-wise obstruction into a single learner-independent hard prior over finite threshold experiments. From this prior we build a hidden-arm kernel: one uniformly selected arm receives distinct records from an external n -sample, while all other arms are simulated independently. Fixing all internal randomness maps an external one-record replacement to zero or one replacement in the input to A , so the kernel inherits the pair $(\varepsilon_0, \delta_m)$ exactly.

A grand collection of iid arm pools couples the constructed input to an ideal iid sample from the uniform arm mixture. The mixture is realized by the minor-table concept, and Bernstein’s inequality bounds the sole overflow event by $e^{-27/2}$. The iid latent experiment vector is independent of the hidden designation, so uniform-arm averaging identifies hidden-arm risk with mixture risk exactly. The PAC guarantee and bounded-loss conversion then give strict prior-average risk below $1/20$, contradicting the hard prior. The last arithmetic step transfers the unchanged δ_m denominator, chooses $c_\delta = a_\delta / [225(1 + \log 15)]$ and $c = a_{\text{th}}/4$, and retains the zero-overflow $k = 1$ specialization. Complete derivations appear in the appendix.

References

- ALON, N., LIVNI, R., MALLIARIS, M. and MORAN, S. (2018). Private PAC learning implies finite Littlestone dimension. [3](#), [8](#), [9](#), [16](#), [18](#), [23](#), [28](#)
- BOUCHERON, S., LUGOSI, G. and MASSART, P. (2013). *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press. [43](#)
- BUN, M., NISSIM, K., STEMMER, U. and VADHAN, S. (2015). Differentially private release and learning of threshold functions. [3](#), [5](#)
- ERDŐS, P. and RADO, R. (1952). Combinatorial theorems on classifications of subsets of a given set. *Proceedings of the London Mathematical Society* **s3-2** 417–439. [3](#), [9](#)
- HOEFFDING, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association* **58** 13–30. [23](#)
- VON NEUMANN, J. (1928). Zur theorie der gesellschaftsspiele. *Mathematische Annalen* **100** 295–320. [32](#)

A Proof Details

A.1 Finite empirical wrapper

Fix integers $n \geq 1$ and $N \geq 2$, and put $M = 9n$. The finite experiment set is

$$\mathcal{E}_{n,N} = [N] \times [N]^M.$$

For $e = (t, U)$, where $U = (u_1, \dots, u_M)$, let Q_e be uniform on the labeled multiset $\{(u_s, \tau_t(u_s)) : s \in [M]\}$. For a kernel $B : ([N] \times \{0, 1\})^n \rightsquigarrow \{0, 1\}^{[N]}$, write

$$\mathcal{R}_n(B, e) = \mathbb{E}_{S \sim Q_e^n, g \sim B(S)} L_{Q_e}(g).$$

Lemma A.1 (With-replacement secrecy of the sample). *Let K be (ε, δ) -differentially private on ordered databases of size r under replacement adjacency. If $s \geq 2r$ and one draws r indices independently and uniformly with replacement from an ordered s -row database before running K , then, for $0 < \varepsilon \leq 1$, the resulting wrapper is*

$$\left(6\varepsilon \frac{r}{s}, e^{6\varepsilon r/s} \frac{4r}{s} \delta\right)\text{-differentially private.} \quad (\text{C1})$$

The statement includes every multiplicity with which the changed row can be selected.

Proof. This is the corrected secrecy-of-the-sample lemma of [Bun et al. \(2015\)](#), stated for ordered replacement adjacency. Its proof conditions on the binomial multiplicity of the changed row and sums over all multiplicities, so no additional group-privacy argument is needed. We use it only with $r = n$ and $s = 9n$. $\square$

Lemma A.2 (Finite coordinate restriction). *Under the coordinate-measurability convention in Section 1, let B^{arb} be an arbitrary-output kernel on ordered samples in $([N] \times \{0, 1\})^n$, and suppose as a local conditional hypothesis that B^{arb} is (ε, δ) -DP under replacement adjacency. Then the coordinate map*

$$\rho(h) = (h(1), \dots, h(N)) \in \{0, 1\}^N \quad (1)$$

is measurable, the pushforward kernel $B := \rho_{\#} B^{\text{arb}}$ is (ε, δ) -DP, and under the canonical coupling $g = \rho(h)$,

$$L_Q(g) = L_Q(h) \quad (2)$$

for every labeled distribution Q supported on $[N] \times \{0, 1\}$. The codomain is the full cube of arbitrary bit vectors, not the threshold subclass.

Proof. Give $\{0, 1\}^N$ its discrete sigma-algebra. For every $v \in \{0, 1\}^N$,

$$\rho^{-1}(\{v\}) = \bigcap_{q=1}^N \{h : h(q) = v_q\}. \quad (3)$$

Every set on the right is measurable by coordinate measurability, and the intersection is finite. Hence ρ is measurable.

For adjacent samples S, S' and every event $E \subseteq \{0, 1\}^N$, the two DP inequalities for B^{arb} applied to the measurable event $\rho^{-1}(E)$ give

$$B(S)(E) = B^{\text{arb}}(S)(\rho^{-1}(E)) \leq e^\varepsilon B^{\text{arb}}(S')(\rho^{-1}(E)) + \delta = e^\varepsilon B(S')(E) + \delta, \quad (4)$$

and the same inequality with S, S' exchanged. Thus B is (ε, δ) -DP.

Finally, under $g = \rho(h)$, one has $g(q) = h(q)$ for every $q \in [N]$. Therefore the error indicators agree pointwise on every (q, y) , which proves (2). No threshold shape has been imposed: $\{0, 1\}^N$ contains every binary function on $[N]$. Conversely every such vector canonically defines an arbitrary hypothesis on $[N]$, so the re-encoding loses no finite-game output. $\square$

Proposition A.1 (Private with-replacement empirical wrapper). *Under Lemma A.2 and the local conditional hypothesis that B is (ε, δ) -DP on ordered n -samples, let $M = 9n$, $0 < \varepsilon \leq 1$, and $\delta \geq 0$. For an ordered labeled database*

$$D = (d_1, \dots, d_M) \in ([N] \times \{0, 1\})^M,$$

define $W_B(D)$ by drawing $K_1, \dots, K_n$ independently and uniformly from $[M]$, drawing

$$g \sim B(d_{K_1}, \dots, d_{K_n}),$$

and outputting g . Then W_B is (ε, δ) -DP under replacement adjacency on ordered M -row databases. This conclusion includes all repeated-index outcomes.

Proof. The wrapper is a well-defined kernel: for every output event E ,

$$W_B(D)(E) = \frac{1}{M^n} \sum_{(k_1, \dots, k_n) \in [M]^n} B(d_{k_1}, \dots, d_{k_n})(E), \quad (5)$$

a finite mixture of measurable kernel evaluations.

Apply Lemma A.1 with source mechanism sample size $r = n$ and source database size $s = M = 9n$. The size condition is exact:

$$s = M = 9n \geq 2n = 2r. \quad (6)$$

It gives the valid privacy parameters

$$\tilde{\varepsilon} = 6\varepsilon \frac{n}{9n} = \frac{2}{3}\varepsilon, \quad \tilde{\delta} = e^{6\varepsilon n/(9n)} \frac{4n}{9n} \delta = \frac{4}{9} e^{2\varepsilon/3} \delta. \quad (7)$$

For $0 < \varepsilon \leq 1$,

$$\tilde{\varepsilon} = \frac{2}{3}\varepsilon \leq \varepsilon. \quad (8)$$

For completeness,

$$e = 2 + \sum_{j=2}^{\infty} \frac{1}{j!} < 2 + \sum_{j=2}^{\infty} \frac{1}{2^{j-1}} = 3, \quad (9a)$$

where $j! \geq 2^{j-1}$ for $j \geq 2$, with strict inequality for some terms. Hence

$$e^{2\varepsilon/3} \leq e^{2/3} < 3^{2/3} < \frac{9}{4},$$

where the final strict inequality follows by cubing, since $9 < 729/64$. Consequently

$$\tilde{\delta} \leq \frac{4}{9} e^{2/3} \delta < \delta \quad \text{when } \delta > 0, \quad \tilde{\delta} = 0 = \delta \quad \text{when } \delta = 0. \quad (9)$$

Monotonicity of the two DP inequalities in both privacy parameters now yields (ε, δ) -DP.

For the repeated-row issue, let D, D' differ at row j , and let

$$L = |\{i \in [n] : K_i = j\}|.$$

Lemma A.1 conditions on every possible multiplicity $L = k$ and sums its binomial probability. Thus (7) already accounts for the cases $k \geq 2$; the proof does not replace the selected samples by adjacent samples on those outcomes, nor does the present argument invoke group privacy or composition outside the cited calculation. $\square$

Lemma A.3 (Zero-residual empirical-risk identity). *Under the definitions of Q_e and $\mathcal{R}_n$ at the beginning of this subsection, let $M = 9n$, let B be any $\{0, 1\}^N$ -valued n -sample kernel, and let W_B be the wrapper of Proposition A.1. For every $t \in [N]$ and every ordered tuple $U = (u_1, \dots, u_M) \in [N]^M$, define*

$$D_{t,U} = ((u_1, \tau_t(u_1)), \dots, (u_M, \tau_t(u_M))). \quad (10)$$

Write its empirical zero-one loss as

$$L_{D_{t,U}}(g) := \frac{1}{M} \sum_{s=1}^M \mathbf{1}\{g(u_s) \neq \tau_t(u_s)\}. \quad (10a)$$

Then

$$\mathbb{E}_{g \sim W_B(D_{t,U})} \left[\frac{1}{M} \sum_{s=1}^M \mathbf{1}\{g(u_s) \neq \tau_t(u_s)\} \right] = \mathcal{R}_n(B, (t, U)). \quad (11)$$

The equality holds with zero residual even when U or the sampled indices contain repetitions.

Proof. By definition, $Q_{(t,U)}$ is uniform on the labeled multiset of the M rows of $D_{t,U}$. Therefore, for every fixed $g \in \{0, 1\}^N$,

$$L_{Q_{(t,U)}}(g) = \frac{1}{M} \sum_{s=1}^M \mathbf{1}\{g(u_s) \neq \tau_t(u_s)\}. \quad (12)$$

This is an identity of finite sums and counts row multiplicities on both sides.

The wrapper indices $K_1, \dots, K_n$ are iid uniform on $[M]$. Hence the selected labeled records

$$S = (D_{t,U}[K_1], \dots, D_{t,U}[K_n])$$

have exactly the product law $Q_{(t,U)}^n$. This remains exact when distinct indices hold equal row values, because $Q_{(t,U)}$ is a distribution on the multiset of row occurrences. Taking expectation in (12), first over g conditionally on S and then over the iid indices, gives

$$\begin{aligned} \mathbb{E}_{g \sim W_B(D_{t,U})} \left[\frac{1}{M} \sum_{s=1}^M \mathbf{1}\{g(u_s) \neq \tau_t(u_s)\} \right] &= \mathbb{E}_{S \sim Q_{(t,U)}^n, g \sim B(S)} L_{Q_{(t,U)}}(g) \\ &= \mathcal{R}_n(B, (t, U)), \end{aligned} \quad (13)$$

which is the claimed exact equality. No concentration, generalization, or population-to-empirical comparison is used. $\square$

Proposition A.2 (Threshold-convention equivalence). *Under the definition $\tau_t(q) = \mathbf{1}\{q \leq t\}$, define*

$$\psi(q) = N + 1 - q, \quad \lambda(1) = +1, \quad \lambda(0) = -1. \quad (14)$$

Then the recordwise bijection

$$(q, y) \mapsto (\psi(q), \lambda(y)) \quad (15)$$

maps every lower-oriented threshold-labeled tuple to a tuple labeled by the upper-oriented threshold convention of Alon et al. (2018), preserves ordered-tuple replacement adjacency and uniform empirical distributions, and induces a bijection of the full arbitrary-output cubes that preserves zero-one loss exactly.

Proof. Put $\theta = N + 1 - t$. For every $q \in [N]$,

$$\lambda(\tau_t(q)) = +1 \iff q \leq t \iff N + 1 - q \geq N + 1 - t \iff \psi(q) \geq \theta. \quad (16)$$

Thus a lower-oriented threshold becomes an upper-oriented source threshold. In particular, after sorting the source coordinates increasingly, negative labels precede positive labels, exactly as stated before the homogeneous-set definition of Alon et al. (2018).

Applying (15) row by row leaves tuple positions unchanged and is a bijection on the record space. Two ordered tuples therefore differ in zero or one positions before the map if and only if they do so after the map. Sampling a uniform row occurrence before applying the map is the same as applying the map and then sampling a uniform row occurrence, so uniform empirical distributions, including multiplicities, are preserved.

For $g \in \{0, 1\}^N$, define the source-sign hypothesis $\Gamma(g)$ by

$$\Gamma(g)(\psi(q)) = \lambda(g(q)). \quad (17)$$

This is a bijection between the full binary function cubes. For every branch labeled record (q, y) ,

$$\mathbf{1}\{g(q) \neq y\} = \mathbf{1}\{\Gamma(g)(\psi(q)) \neq \lambda(y)\}. \quad (18)$$

Hence empirical zero-one loss is exactly invariant. Because Γ maps every bit vector, rather than only threshold vectors, the convention change preserves arbitrary improper outputs. $\square$

Proposition A.3 (Finite empirical-wrapper interface). *For every $n \geq 1$, $N \geq 2$, $0 < \varepsilon \leq 1$, and $\delta \geq 0$, every arbitrary-output (ε, δ) -private n -sample threshold kernel admits a measurable restriction B to the full cube $\{0, 1\}^{[N]}$. With $M = 9n$, its iid with-replacement empirical wrapper W_B is (ε, δ) -private and, for every $e = (t, U)$,*

$$\mathbb{E}_{g \sim W_B(D_{t,U})} L_{D_{t,U}}(g) = \mathcal{R}_n(B, e).$$

The equality includes repeated rows and indices, and the order reversal $q \mapsto N + 1 - q$ preserves realizability, zero-one loss, replacement adjacency, empirical distributions, and the full arbitrary-output cube.

Proof. Start with any candidate arbitrary-output (ε, δ) -DP n -sample threshold-domain kernel, with $0 < \varepsilon \leq 1$. Lemma A.2 measurably re-encodes its predictions on $[N]$ as a kernel B taking values in the full finite cube $\{0, 1\}^N$, with no privacy or risk change and no properness restriction.

Set $M = 9n$ and form W_B by iid uniform with-replacement row sampling. Proposition A.1, using the fully instantiated corrected secrecy lemma, gives the raw certificate

$$\left(\frac{2}{3}\varepsilon, \frac{4}{9}e^{2\varepsilon/3}\delta \right),$$

which is dominated by (ε, δ) throughout $0 < \varepsilon \leq 1$. The corrected multiplicity calculation includes repeated selection of the changed database row, so there is no group-privacy or composition residual.

For every $e = (t, U) \in \mathcal{E}_{n,N}$, Lemma A.3 gives

$$\mathbb{E}_{g \sim W_B(D_{t,U})} L_{D_{t,U}}(g) = \mathcal{R}_n(B, e) \quad (19)$$

exactly. Lemma A.2 identifies the right-hand side with the risk of the original arbitrary-output kernel, and Proposition A.2 identifies both sides with the empirical-threshold interface of Alon et al. (2018) after a loss- and adjacency-preserving bijection.

Thus the conclusion holds for every candidate kernel and every $9n$ -tuple U , uniformly through the endpoint $\varepsilon = 1$, without using a hard prior or any additional assumption. $\square$

A.2 Homogeneity and endpoint gap

In this subsection B is a full-cube (ε, δ) -private n -sample kernel, $M = 9n$, and the empirical wrapper of Proposition A.3 satisfies

$$\mathcal{R}_n(B, e) \leq \frac{1}{20} \quad \text{for every } e \in \mathcal{E}_{n,N}. \quad (\text{A1})$$

If D is any ordered threshold-labeled M -tuple and e_D is its corresponding finite experiment, the empirical wrapper W_B from Proposition A.3 obeys the exact identity

$$\mathbb{E}_{g \sim W_B(D)} L_D(g) = \mathcal{R}_n(B, e_D). \quad (\text{C1})$$

Define

$$g_* := \frac{9}{10e} - \frac{e}{10}, \quad g_{\text{odd}} := \frac{71}{80e} - \frac{9e}{80}, \quad g_{\text{gap}} := \min\{g_*, g_{\text{odd}}\},$$

and impose the local scalar condition

$$0 \leq \delta \leq \frac{a_{\delta, \text{end}}}{n^2 \log(en)}, \quad a_{\delta, \text{end}} := \frac{g_{\text{gap}}}{8(1 + e^{-1})}. \quad (\text{A2})$$

The positivity of these constants is proved below, before the condition is used.

Theorem A.1 (Finite Ramsey theorem). *Define $\text{twr}_1(u) = u$ and $\text{twr}_{r+1}(u) = 2^{\text{twr}_r(u)}$. If integers $s > t \geq 2$ and $q \geq 2$ satisfy*

$$N \geq \text{twr}_t(3sq \log_2 q), \quad (\text{C2})$$

then every coloring of the t -subsets of an N -point set with at most q colors has a monochromatic s -subset.

Proof. This finite form, established by Erdős and Rado (1952), is used in the homogeneous-set proof of Alon et al. (2018). The application below sets $t = M + 1$ and verifies the tower inequality explicitly. $\square$

Lemma A.4 (Upper-oriented symmetric empirical wrapper). *Under Proposition A.3, let B be a full-cube (ε, δ) -DP n -sample kernel, let $M = 9n$, and let W_B be its empirical wrapper. Define*

$$\psi(q) = N + 1 - q. \quad (1)$$

Conjugating input records and output coordinates by ψ gives an upper-oriented wrapper W on M -row tuples over $[N] \times \{0, 1\}$. Then W is (ε, δ) -DP, has exactly the same empirical zero-one loss as W_B , takes values in the full cube $\{0, 1\}^{[N]}$, and for every row permutation π ,

$$W(D) \stackrel{\text{law}}{=} W(\pi D). \quad (2)$$

Consequently, under (A1), every upper-oriented threshold-realizable tuple D satisfies

$$\mathbb{E}_{v \sim W(D)} L_D(v) \leq \frac{1}{20}. \quad (3)$$

Proof. Proposition A.2 maps q to $x = \psi(q)$ and maps the branch target to

$$\sigma_{\vartheta}(x) = \mathbf{1}\{x \geq \vartheta\}, \quad \vartheta = N + 1 - t. \quad (4)$$

The record map is bijective and position preserving, and the output map is the coordinate relabeling

$$v(x) = g(\psi^{-1}(x)). \quad (5)$$

Thus DP, adjacency, zero-one loss, and the full arbitrary-output cube are preserved exactly. Equation (3) follows from (A1) and Lemma A.3.

It remains to prove (2). The wrapper has the finite-mixture formula

$$W_B(D)(E) = \frac{1}{M^n} \sum_{(k_1, \dots, k_n) \in [M]^n} B(d_{k_1}, \dots, d_{k_n})(E). \quad (6)$$

If π permutes the rows, the map

$$(k_1, \dots, k_n) \mapsto (\pi(k_1), \dots, \pi(k_n))$$

is a bijection of $[M]^n$. Reindexing the sum in (6) proves $W_B(D) = W_B(\pi D)$ as output laws, even when B itself is order sensitive. Conjugating by (1) and (5) preserves this equality and proves (2). Therefore a tuple obtained by one record replacement may be sorted before applying homogeneity: privacy compares the original tuple with the unsorted adjacent tuple, while (2) identifies the latter's output law with that of its sorted version. This sorting introduces neither a second replacement nor a composition charge. □

Lemma A.5 (Universal numerical endpoint gap). *The constants*

$$g_* := \frac{9}{10e} - \frac{e}{10}, \quad g_{\text{odd}} := \frac{71}{80e} - \frac{9e}{80}, \quad g_{\text{gap}} := \min\{g_*, g_{\text{odd}}\} \quad (\text{N1})$$

are universal and strictly positive. Moreover $g_{\text{gap}} < 1$.

Proof. For every integer $r \geq 3$,

$$r! \geq 6 \cdot 3^{r-3}.$$

The inequality is strict for at least one $r \geq 4$, so

$$e = 1 + 1 + \frac{1}{2} + \sum_{r=3}^{\infty} \frac{1}{r!} < \frac{5}{2} + \frac{1}{6} \sum_{u=0}^{\infty} 3^{-u} = \frac{11}{4}. \quad (\text{N2})$$

In particular $e < 3$, and therefore

$$g_* = \frac{9 - e^2}{10e} > 0. \quad (\text{N3})$$

Also

$$9e^2 < 9 \left(\frac{11}{4} \right)^2 = \frac{1089}{16} < 71, \quad (\text{N4})$$

which gives

$$g_{\text{odd}} = \frac{71 - 9e^2}{80e} > 0. \quad (\text{N5})$$

Thus $g_{\text{gap}} > 0$. Finally, both $g_* < 9/(10e) < 1$ and $g_{\text{odd}} < 71/(80e) < 1$, so $g_{\text{gap}} < 1$. □

Proposition A.4 (Finite nearly balanced homogeneity coloring). *Under Lemma A.4, let $M \geq 9$, set*

$$s_- := \left\lfloor \frac{M}{2} \right\rfloor, \quad s_+ := M - s_-, \quad (7)$$

and fix the rank-label template consisting of s_- zeros followed by s_+ ones. For any $\gamma \in (0, 1)$, there is a coloring of the $(M + 1)$ -subsets of $[N]$ using at most

$$q_M(\gamma) = \left(1 + \left\lceil \frac{1}{\gamma} \right\rceil\right)^{M+1} \quad (8)$$

colors such that every monochromatic set H has a list

$$(p_0, \dots, p_M) \in [0, 1]^{M+1} \quad (9)$$

with the following property. If S is any increasing M -point sample from H carrying the fixed template, $x \in H \setminus S_X$, and

$$r = \text{ord}_S(x) := |\{z \in S_X : z \leq x\}| \in \{0, \dots, M\}, \quad (10)$$

then, writing

$$a_S(x) := \Pr_{v \sim \mathbf{W}(S)}[v(x) = 1], \quad (11)$$

one has

$$|a_S(x) - p_r| \leq \gamma. \quad (12)$$

This conclusion uses only coordinate marginals and imposes no threshold shape or properness on the output.

Proof. Put

$$L = \left\lceil \frac{1}{\gamma} \right\rceil, \quad \mathcal{G}_L = \left\{0, \frac{1}{L}, \dots, 1\right\}. \quad (13)$$

Round each $u \in [0, 1]$ to a nearest element $Q_\gamma(u)$ of $\mathcal{G}_L$, breaking ties toward the smaller grid point. Since the grid spacing is $1/L \leq \gamma$,

$$|Q_\gamma(u) - u| \leq \frac{1}{2L} \leq \gamma. \quad (14)$$

For an ordered $(M + 1)$ -subset

$$D = \{x_1 < \dots < x_{M+1}\},$$

let S^{-i} be the increasing sample on $D \setminus \{x_i\}$ with the fixed rank-label template (7). Color D by

$$\text{col}(D) = (Q_\gamma(a_{S^{-1}}(x_1)), \dots, Q_\gamma(a_{S^{-(M+1)}}(x_{M+1}))). \quad (15)$$

Every coordinate has $L + 1 = 1 + \text{ceil}(1/\gamma)$ possibilities, proving (8).

Suppose all $(M + 1)$ -subsets of H have the same color $(c_1, \dots, c_{M+1})$. Define

$$p_r = c_{r+1}, \quad 0 \leq r \leq M. \quad (16)$$

Given S, x in the statement, the set $D = S_X \cup \{x\}$ has x in position $r + 1$, and deleting x from D gives exactly the sorted fixed-template sample S . Thus the $(r + 1)$ -st coordinate of (15) is $Q_\gamma(a_S(x)) = p_r$. Equation (14) proves (12).

The cases $r = 0$ and $r = M$ are literal coloring coordinates: the queried point lies before all or after all sample points. No point outside H , no arithmetic successor, and no output-shape assumption is used. □

Proposition A.5 (Finite Ramsey extraction with adjusted color constants). *Under Lemma A.5 and Proposition A.4, take*

$$\gamma := \frac{g_{\text{gap}}}{100M}, \quad q_M := q_M(\gamma). \quad (17)$$

There is a universal constant $C_R > 0$, depending only on the numerical constant g_{gap} , with the following property. Whenever $N \geq M + 1$ and the base-two iterated logarithm

$$\ell_M(N) := \log_2^{(M)} N \quad (18)$$

is defined and at least one, the coloring in Proposition A.4 has a monochromatic set H satisfying

$$|H| \geq M + 1, \quad |H| \geq \frac{\ell_M(N)}{\exp(C_R M \log(eM))}. \quad (19)$$

The exact finite bound before universal simplification is

$$|H| \geq \frac{\ell_M(N)}{6q_M \log_2 q_M}. \quad (20)$$

Proof. Lemma A.5 gives $0 < g_{\text{gap}} < 1$. Hence $\gamma \in (0, 1)$ for $M \geq 9$, and $q_M \geq 2$ is an integer.

Set

$$x_R := \frac{\ell_M(N)}{3q_M \log_2 q_M}, \quad s_R := \lfloor x_R \rfloor. \quad (21)$$

If $s_R \geq M + 2$, instantiate Theorem A.1 with

$$t = M + 1, \quad q = q_M, \quad s = s_R.$$

The required strict inequality $s > t$ holds. Moreover,

$$3s_R q_M \log_2 q_M \leq \ell_M(N). \quad (22)$$

By the definitions of the tower and iterated logarithm,

$$\text{twr}_{M+1}(\ell_M(N)) = N. \quad (23)$$

Monotonicity of the tower, (22), and (23) verify the cited theorem's size hypothesis, so a monochromatic set of size s_R exists.

If $s_R \leq M + 1$, any $(M + 1)$ -subset is monochromatic because it has only one $(M + 1)$ -subset to color. Such a set exists because $N \geq M + 1$. In both branches,

$$|H| \geq \frac{x_R}{2} = \frac{\ell_M(N)}{6q_M \log_2 q_M}. \quad (24)$$

Indeed, in the first branch $\lfloor x_R \rfloor \geq x_R/2$ since $x_R \geq M + 2 \geq 2$; in the second branch $x_R < M + 2 \leq 2(M + 1)$ and $|H| = M + 1 \geq x_R/2$. This proves (20), including the finite rounding case omitted by asymptotic notation.

It remains to bound the color denominator. Define the numerical constant

$$C_g := 2 + \frac{100}{g_{\text{gap}}}. \quad (25)$$

Since $M \geq 1$,

$$1 + \left\lceil \frac{1}{\gamma} \right\rceil = 1 + \left\lceil \frac{100M}{g_{\text{gap}}} \right\rceil \leq C_g M. \quad (26)$$

Therefore

$$q_M \leq (C_g M)^{M+1}, \quad (27)$$

and, with $A_g := 1 + \log C_g$,

$$\log q_M \leq (M+1)(\log C_g + \log M) \leq 2A_g M \log(eM). \quad (28)$$

Because $q_M \geq 2$, $\log_2 q_M \leq q_M$, and hence

$$\begin{aligned} \log(6q_M \log_2 q_M) &\leq \log 6 + 2 \log q_M \\ &\leq (\log 6 + 4A_g) M \log(eM). \end{aligned} \quad (29)$$

Thus the explicit universal choice

$$C_R := \log 6 + 4(1 + \log C_g) \quad (30)$$

turns (20) into (19). Every constant in (19) is therefore numerical and learner independent.

For the finite boundary $N \leq M$, the domain itself has size at most M ; the $(M+1)$ -subset coloring is empty and no off-sample rank list is needed. The nonvacuous Ramsey branch used below is precisely the nonvacuous $N \geq M+1$ branch proved above. $\square$

Lemma A.6 (Parity-uniform side marginals at expected loss one twentieth). *Under Lemma A.4 and (A1), let*

$$S = ((z_1, 0), \dots, (z_{s_-}, 0), (z_{s_-+1}, 1), \dots, (z_M, 1)) \quad (32)$$

be any increasing fixed-template sample on distinct points. Then there are indices

$$\ell \in \{1, \dots, s_-\}, \quad j \in \{s_- + 1, \dots, M\} \quad (33)$$

such that

$$a_S(z_\ell) \leq \theta_M, \quad a_S(z_j) \geq 1 - \theta_M, \quad (34)$$

where

$$\theta_M = \begin{cases} 1/10, & M \text{ even}, \\ 9/80, & M \text{ odd}. \end{cases} \quad (35)$$

The odd bound holds for every odd $M \geq 9$.

Proof. The expected empirical error of $\mathbf{W}$ on S is

$$\frac{1}{M} \left(\sum_{i=1}^{s_-} a_S(z_i) + \sum_{i=s_-+1}^M (1 - a_S(z_i)) \right) \leq \frac{1}{20} \quad (36)$$

by (3). Both sums are nonnegative, so

$$\min_{i \leq s_-} a_S(z_i) \leq \frac{M}{20s_-}, \quad \min_{i > s_-} (1 - a_S(z_i)) \leq \frac{M}{20s_+}. \quad (37)$$

If M is even, both side sizes are $M/2$, and both right-hand sides equal $1/10$.

If $M = 2r + 1$ is odd, then $r = s_- \geq 4$ and $s_+ = r + 1$. Thus

$$\frac{M}{20s_-} = \frac{2r+1}{20r} = \frac{1}{10} + \frac{1}{20r} \leq \frac{9}{80}, \quad (38)$$

while

$$\frac{M}{20s_+} = \frac{2r+1}{20(r+1)} < \frac{1}{10} < \frac{9}{80}. \quad (39)$$

Choosing minimizers in (37) proves (34). At the smallest admitted odd size $M = 9$, (38) is exactly $9/80$, so no asymptotic parity claim is being used. $\square$

Proposition A.6 (Endpoint-uniform homogeneous-list gap). *Under Lemma A.5, Lemma A.4, Proposition A.4, Lemma A.6, and (A2), let H be any monochromatic set with $|H| \geq M + 1$, let $(p_0, \dots, p_M)$ be its homogeneous list at mesh $\gamma = g_{\text{gap}}/(100M)$, and suppose (A1) holds. Then there are ranks*

$$0 \leq r_- < r_+ \leq M \quad (40)$$

such that, uniformly for every $0 < \varepsilon \leq 1$,

$$p_{r_+} - p_{r_-} \geq g_{\text{gap}} - (1 + e^{-1})\delta - 2\gamma \geq \frac{3g_{\text{gap}}}{4}. \quad (41)$$

Both constants in (N1) are strictly positive; in particular the gap at the privacy endpoint $\varepsilon = 1$ is proved rather than assumed.

Proof. 1. One unused point and the two realizable replacements.

Choose any ordered $(M + 1)$ -subset

$$h_1 < \dots < h_{M+1} \quad (42)$$

of H , and leave the central boundary point

$$x_* := h_{s_-+1} \quad (43)$$

out of the base sample. Write the remaining sorted points as

$$z_i = \begin{cases} h_i, & i \leq s_-, \\ h_{i+1}, & i > s_-. \end{cases} \quad (44)$$

Give $z_1, \dots, z_{s_-}$ label zero and the remaining points label one. This is the upper-oriented threshold sample whose first positive point is h_{s_-+2} . Lemma A.6 supplies ℓ, j satisfying (33)-(34).

For the low-side comparison, replace the record $(z_\ell, 0)$ by $(x_*, 0)$. After sorting, the resulting sample S^- still has exactly s_- zeros followed by s_+ ones and is realized by the upper threshold beginning at h_{s_-+2} . The removed point has rank

$$\text{ord}_{S^-}(z_\ell) = \ell - 1. \quad (45)$$

For the high-side comparison, replace $(z_j, 1)$ by $(x_*, 1)$. After sorting, the resulting sample S^+ again has the fixed template and is realized by the upper threshold beginning at $x_* = h_{s_-+1}$. The removed point has rank

$$\text{ord}_{S^+}(z_j) = j. \quad (46)$$

The same unused domain point may carry different labels in the two separate neighboring samples because each sample is independently threshold realizable; no theorem assumption identifies their latent targets.

The direct replacements are one-record replacements of the ordered base tuple but need not remain sorted. Lemma A.4 identifies the law on each unsorted adjacent tuple with the law on its

sorted version. Hence (45)-(46) use exactly one privacy comparison each. The construction uses only the next available point in the finite ordered set (42); it never assumes that an arithmetic successor such as $x + 1$ belongs to the domain. It is therefore valid for arbitrarily sparse H .

2. DP transfer and homogeneity.

Set

$$r_- := \ell - 1, \quad r_+ := j. \quad (47)$$

Since $\ell \leq s_- < j$, (40) holds. Apply the DP upper inequality to the event that the output predicts one at z_ℓ :

$$a_{S^-}(z_\ell) \leq e^\varepsilon a_S(z_\ell) + \delta \leq e^\varepsilon \theta_M + \delta. \quad (48)$$

By homogeneity and (45),

$$p_{r_-} \leq e^\varepsilon \theta_M + \delta + \gamma. \quad (49)$$

For z_j , use the DP inequality in the direction from the base sample to its replacement:

$$a_S(z_j) \leq e^\varepsilon a_{S^+}(z_j) + \delta.$$

Thus

$$a_{S^+}(z_j) \geq e^{-\varepsilon}(a_S(z_j) - \delta) \geq e^{-\varepsilon}(1 - \theta_M - \delta), \quad (50)$$

and homogeneity with (46) gives

$$p_{r_+} \geq e^{-\varepsilon}(1 - \theta_M - \delta) - \gamma. \quad (51)$$

Subtracting (49) from (51),

$$p_{r_+} - p_{r_-} \geq e^{-\varepsilon}(1 - \theta_M - \delta) - e^\varepsilon \theta_M - \delta - 2\gamma. \quad (52)$$

3. The privacy endpoint and its defects.

The delta condition (A2), $n^2 \log(en) \geq 1$, and $g_{\text{gap}} < 1$ imply $\delta < 1/8 < 1 - \theta_M$. Therefore the derivative of

$$F_M(u) := e^{-u}(1 - \theta_M - \delta) - e^u \theta_M - \delta \quad (53)$$

is

$$F'_M(u) = -e^{-u}(1 - \theta_M - \delta) - e^u \theta_M < 0. \quad (54)$$

Thus the worst point of the whole expression, including its delta coefficient, is $u = 1$; one must not minimize the multiplicative terms and the additive terms separately. Equations (52)-(54) yield

$$p_{r_+} - p_{r_-} \geq e^{-1}(1 - \theta_M) - e \theta_M - (1 + e^{-1})\delta - 2\gamma. \quad (55)$$

For even M , the first two terms in (55) equal

$$g_* = \frac{9}{10e} - \frac{e}{10} = \frac{9 - e^2}{10e}. \quad (56)$$

For odd M , they equal

$$g_{\text{odd}} = \frac{71}{80e} - \frac{9e}{80} = \frac{71 - 9e^2}{80e}. \quad (57)$$

Lemma A.5 proves both expressions strictly positive without decimal approximation. Therefore each parity-specific first term in (55) is at least the universal g_{gap} , and the claimed $\varepsilon = 1$ endpoint gap is established rather than postulated.

Using (A2) and $n^2 \log(en) \geq 1$,

$$(1 + e^{-1})\delta \leq \frac{g_{\text{gap}}}{8}. \quad (61)$$

Also $M \geq 9$ and (17) imply

$$2\gamma = \frac{g_{\text{gap}}}{50M} \leq \frac{g_{\text{gap}}}{450}. \quad (62)$$

Substituting (56)-(57) and (61)-(62) into (55), and using Lemma A.5, gives

$$p_{r_+} - p_{r_-} \geq \left(1 - \frac{1}{8} - \frac{1}{450}\right) g_{\text{gap}} = \frac{1571}{1800} g_{\text{gap}} > \frac{3}{4} g_{\text{gap}}, \quad (63)$$

which proves (41). At $\delta = 0$, the defect in (61) vanishes, so the boundary only improves.

Finally, (45) permits $r_- = 0$ when $\ell = 1$, and (46) permits $r_+ = M$ when $j = M$. Thus both one-sided endpoints of the homogeneous list are genuine cases of the same construction, with no phantom point beyond H . $\square$

Proposition A.7 (Adjacent rise from the endpoint-uniform gap). *Under Proposition A.6, some $i \in [M]$ satisfies*

$$p_i - p_{i-1} \geq \frac{3g_{\text{gap}}}{4M}. \quad (64)$$

Proof. For the ranks $r_- < r_+$ from Proposition A.6,

$$p_{r_+} - p_{r_-} = \sum_{i=r_-+1}^{r_+} (p_i - p_{i-1}). \quad (65)$$

There are $r_+ - r_- \leq M$ summands. If every summand were smaller than $3g_{\text{gap}}/(4M)$, then the sum in (65) would be smaller than $3g_{\text{gap}}/4$, contradicting (41). Hence (64) holds for some

$$i \in \{r_- + 1, \dots, r_+\} \subseteq [M].$$

The proof does not assume monotonicity of the list; it uses only the signed total rise (41). The alternatives $i = 1$ and $i = M$ are allowed because Proposition A.6 includes $r_- = 0$ and $r_+ = M$. $\square$

Proposition A.8 (Homogeneous set with an adjacent rise). *Under Proposition A.3, let $M = 9n$, suppose $0 < \varepsilon \leq 1$, and assume (A1)–(A2). In the nonvacuous finite-Ramsey branch $N \geq M + 1$, there are a universal $C_R > 0$, a homogeneous set $H \subseteq [N]$, a list $(p_0, \dots, p_M)$, and $i \in [M]$ such that*

$$|H| \geq M + 1, \quad |H| \geq \frac{\log_2^{(M)} N}{\exp(C_R M \log(eM))}, \quad p_i - p_{i-1} \geq \frac{3g_{\text{gap}}}{4M}.$$

The statement holds for both parities of M and includes $\varepsilon = 1$ and the endpoint indices $i = 1, M$. When $N \leq M$, the finite boundary needs no homogeneous-list construction.

Proof. Fix a candidate n -sample threshold kernel B with $0 < \varepsilon \leq 1$ and assume (A1). Proposition A.3 supplies its full-cube (ε, δ) -DP wrapper on $M = 9n$ rows and the exact equality between wrapper expected empirical loss and $\mathcal{R}_n(B, e)$. Lemma A.4 conjugates this object to the upper-threshold orientation of Alon et al. (2018), preserves arbitrary outputs, privacy, and loss exactly, and proves the row-permutation invariance needed to sort a one-record replacement without changing its law.

Define the universal numerical constants

$$g_* = \frac{9}{10e} - \frac{e}{10}, \quad g_{\text{odd}} = \frac{71}{80e} - \frac{9e}{80}, \quad g_{\text{gap}} = \min\{g_*, g_{\text{odd}}\} > 0, \quad (66)$$

the mesh

$$\gamma = \frac{g_{\text{gap}}}{100M}, \quad (67)$$

and the endpoint delta constant

$$a_{\delta, \text{end}} = \frac{g_{\text{gap}}}{8(1 + e^{-1})} > 0. \quad (68)$$

Proposition A.4 gives a finite coloring with at most

$$q_M = \left(1 + \left\lceil \frac{100M}{g_{\text{gap}}} \right\rceil\right)^{M+1} \quad (69)$$

colors and a homogeneous list on every monochromatic set. Proposition A.5, after independently instantiating Theorem A.1, gives such a set with

$$|H| \geq \frac{\log_2^{(M)} N}{\exp(C_R M \log(eM))} \quad (70)$$

in the nonvacuous finite-Ramsey range, where the explicit C_R in (30) is universal. This is the Ramsey lower bound with the odd-template and finer-mesh color constants fully exposed.

On any $(M + 1)$ points of H , Lemma A.6 converts expected empirical loss at most $1/20$ into side errors $1/10$ for even M and at most $9/80$ for odd $M \geq 9$. Proposition A.6 uses one unused threshold-boundary point and exactly two one-record DP comparisons to obtain ranks $r_- < r_+$ with

$$p_{r_+} - p_{r_-} \geq \frac{3g_{\text{gap}}}{4}. \quad (71)$$

Its calculation minimizes the full privacy expression at $\varepsilon = 1$, proves both constants in (66) positive, and separately controls the approximate-DP and mesh defects. Proposition A.7 then yields

$$p_i - p_{i-1} \geq \frac{3g_{\text{gap}}}{4M} \quad (72)$$

for some $i \in [M]$, including the possible one-sided cases $i = 1$ and $i = M$.

Applying ψ^{-1} to H and to every queried coordinate returns to the lower-oriented threshold coordinates with identical marginals, loss, and replacement adjacency by the dependency. The resulting order is the pullback of the upper-oriented order,

$$q \preceq q' \iff \psi(q) \leq \psi(q'),$$

equivalently the reverse of the natural q -order. With this declared order, ranks, the list indexing, and the sign of (72) are unchanged. If one instead writes the image in increasing natural q -order, one must reverse the list and the adjacent inequality simultaneously. Thus (66), (70), and (72) are on the exact finite-output coordinates and declared order consumed by Proposition A.12; no orientation sign is silently changed, and no properization, surrogate loss, hard prior, or later-step conclusion has entered. This proves the conclusion. $\square$

A.3 Moving-record private family

Retain the reverse order

$$\psi(q) = N + 1 - q, \quad q \preceq q' \iff \psi(q) \leq \psi(q'). \quad (\text{C1})$$

Proposition A.8 supplies a homogeneous set H , a list $(p_0, \dots, p_M)$, and an index $i \in [M]$ such that

$$p_i - p_{i-1} \geq \frac{3g_{\text{gap}}}{4M}. \quad (\text{A1})$$

For every increasing fixed-template sample S from H and every off-sample $z \in H$, its homogeneity conclusion is

$$\left| \Pr_{g \sim \mathcal{W}(S)} \{g(z) = 1\} - p_{\text{ord}_S^{\preceq}(z)} \right| \leq \gamma, \quad \gamma = \frac{g_{\text{gap}}}{100M}, \quad (\text{C2})$$

where $\text{ord}_S^{\preceq}(z) = |\{u \in S_X : u \preceq z\}|$. Define the universal numerical constant

$$a_0 := \frac{g_{\text{gap}}}{100} > 0. \quad (\text{T1})$$

The laws constructed below, with $\eta = a_0/M$, satisfy for every distinct $x, z \in J$,

$$z \prec x \implies \Pr_{v \sim P_x} \{v(z) = 1\} \leq r - \eta, \quad x \prec z \implies \Pr_{v \sim P_x} \{v(z) = 1\} \geq r + \eta. \quad (\text{T2})$$

The construction below is the endpoint-safe current-notation version of the moving-record mechanism in Alon et al. (2018); every rank, adjacency relation, and margin is rederived here.

Proposition A.9 (Fixed-extremes moving-record databases). *Under Proposition A.8, let*

$$H = \{h_1 \prec h_2 \prec \dots \prec h_L\}, \quad L := |H| \geq M + 1, \quad (1)$$

and let $i \in [M]$ satisfy (A1). Put

$$s_- := \left\lfloor \frac{M}{2} \right\rfloor, \quad \lambda_r := \mathbf{1}\{r > s_-\}, \quad r \in [M]. \quad (2)$$

Define the $\preceq$ -interval in H

$$J := \{h_i, h_{i+1}, \dots, h_{L-M+i}\}. \quad (3)$$

For each $x \in J$, define the ordered M -tuple

$$S_x = (d_1(x), \dots, d_M(x)), \quad (4)$$

where

$$d_r(x) = \begin{cases} (h_r, \lambda_r), & r < i, \\ (x, \lambda_i), & r = i, \\ (h_{L-M+r}, \lambda_r), & r > i. \end{cases} \quad (5)$$

Then $|J| = L - M + 1$; every S_x has exactly M distinct domain coordinates, is $\preceq$ -increasing and realizable by a lower-oriented threshold; and, for every distinct $x, x' \in J$, the ordered tuples $S_x, S_{x'}$ differ in exactly the record at position i .

Proof. The left fixed coordinates are

$$h_1, \dots, h_{i-1}, \quad (6)$$

and the right fixed coordinates are

$$h_{L-M+i+1}, \dots, h_L. \quad (7)$$

There are $i - 1$ coordinates in (6) and $M - i$ in (7), hence exactly $M - 1$ fixed coordinates. The remaining consecutive block of H is (3), and

$$|J| = (L - M + i) - i + 1 = L - M + 1 \geq 2. \quad (8)$$

The three index ranges

$$\{1, \dots, i-1\}, \quad \{i, \dots, L-M+i\}, \quad \{L-M+i+1, \dots, L\} \quad (9)$$

are disjoint. Therefore every $x \in J$ is different from all fixed coordinates, the coordinates in S_x are distinct, and (5) is $\preceq$ -increasing with x at the same position i for every choice of x .

The labels in (2) are zero through rank s_- and one from rank $s_- + 1$ onward. Because $\preceq$ is the pullback upper order, $u \prec u'$ means $u > u'$ in the natural present coordinate. If $u_r(x)$ is the domain coordinate of $d_r(x)$, choose

$$t_x := u_{s_-+1}(x) \in [N]. \quad (10)$$

For $r \leq s_-$, natural-order strictness gives $u_r(x) > t_x$, and hence $\tau_{t_x}(u_r(x)) = 0 = \lambda_r$. For $r \geq s_- + 1$, it gives $u_r(x) \leq t_x$, and hence $\tau_{t_x}(u_r(x)) = 1 = \lambda_r$. Thus S_x is exactly threshold-realizable. This argument also covers the possibility that the moving coordinate itself is the first positive point.

For distinct $x, x' \in J$, equations (4)–(5) show that every position except i contains the same labeled record, whereas position i contains (x, λ_i) versus (x', λ_i) . Hence the tuples differ in exactly one replacement. No sorting, permutation, group-privacy comparison, or chain of adjacent databases is used.

At the left endpoint $i = 1$, (6) is empty,

$$J = \{h_1, \dots, h_{L-M+1}\}, \quad (11)$$

and all $M - 1$ fixed records form the suffix (7). At the right endpoint $i = M$, (7) is empty,

$$J = \{h_M, \dots, h_L\}, \quad (12)$$

and all $M - 1$ fixed records form the prefix (6). Thus the construction never calls for h_0 , h_{L+1} , or any ambient arithmetic successor. □

Lemma A.7 (Exact off-diagonal ranks in the moving block). *Under Proposition A.9, for every $x, z \in J$ with $z \neq x$,*

$$\text{ord}_{S_x}^{\prec}(z) = \begin{cases} i-1, & z \prec x, \\ i, & x \prec z. \end{cases} \quad (13)$$

Consequently, $i = 1$ gives the one-sided rank pair $0, 1$, and $i = M$ gives the one-sided rank pair $M - 1, M$, whenever the corresponding side contains a query point.

Proof. Every point of the fixed prefix (6) precedes every point of J , and every point of the fixed suffix (7) follows every point of J .

If $z \prec x$, the only sample coordinates u satisfying $u \preceq z$ are the $i - 1$ fixed prefix coordinates. The moving point x and every fixed suffix coordinate follow z . Thus the rank is $i - 1$.

If $x \prec z$, the $i - 1$ fixed prefix coordinates and the moving coordinate x satisfy $u \preceq z$, while every fixed suffix coordinate follows z . Thus the rank is i .

When $i = 1$, the fixed prefix is empty, so the two possible off-diagonal ranks are literally 0 and 1. When $i = M$, the fixed suffix is empty, so they are literally $M - 1$ and M . If x is the first or last point of J , one of the two implications in (13) has no eligible z and is simply vacuous; no phantom query coordinate is introduced. The diagonal $z = x$ lies in the sample and is outside the homogeneity interface, so no rank-based marginal claim is made there. □

Proposition A.10 (Parameter-preserving arbitrary-output restrictions). *Under the exact (ε, δ) privacy of $\mathcal{W}$ and Proposition A.9, let*

$$\rho_J : \{0, 1\}^{[N]} \longrightarrow \{0, 1\}^J, \quad \rho_J(g) = g|_J, \quad (14)$$

and define

$$P_x := \text{Law}(\rho_J(g)), \quad g \sim \mathcal{W}(S_x), \quad x \in J. \quad (15)$$

Then, for every distinct $x, x' \in J$ and every event $E \subseteq \{0, 1\}^J$,

$$P_x(E) \leq e^\varepsilon P_{x'}(E) + \delta, \quad P_{x'}(E) \leq e^\varepsilon P_x(E) + \delta. \quad (16)$$

The laws in (15) may be supported on arbitrary bit vectors; no threshold shape, monotonicity, properness, representation, or computational restriction is imposed.

Proof. The finite-coordinate map ρ_J is measurable by the coordinate-measurability convention in Section 1, and it is the same deterministic map for every x . For distinct x, x' , Proposition A.9 shows that $S_x, S'_{x'}$ are replacement adjacent by exactly one record. Applying the DP inequality to the event $\rho_J^{-1}(E)$ gives

$$\begin{aligned} P_x(E) &= \Pr_{g \sim \mathcal{W}(S_x)}[g \in \rho_J^{-1}(E)] \\ &\leq e^\varepsilon \Pr_{g \sim \mathcal{W}(S_{x'})}[g \in \rho_J^{-1}(E)] + \delta \\ &= e^\varepsilon P_{x'}(E) + \delta. \end{aligned} \quad (17)$$

Swapping x, x' gives the second inequality in (16). There is one invocation of privacy in each direction and no path through intermediate databases, so neither ε nor δ is composed. Restriction is postprocessing of the full arbitrary-output cube, so it adds no output-shape condition. $\square$

Proposition A.11 (Pullback-oriented coordinate separation). *Under the homogeneity estimate (C2), the adjacent rise (A1), Proposition A.10, and Lemma A.7, define*

$$\Delta_i := p_i - p_{i-1}, \quad r := \frac{p_i + p_{i-1}}{2}, \quad a_0 := \frac{g_{\text{gap}}}{100}, \quad \eta := \frac{a_0}{M}. \quad (18)$$

Then $r \in (\eta, 1 - \eta)$, and for every $x, z \in J$ with $z \neq x$, the laws (15) satisfy (T2). If

$$J = \{j_1 \prec \cdots \prec j_K\}, \quad K = |J|, \quad (19)$$

define the order isomorphism and common coordinate relabeling by

$$\iota(j_\ell) = \ell, \quad (R_\iota v)(s) = v(j_s), \quad (19a)$$

and define

$$\tilde{P}_\ell := (R_\iota)_\# P_{j_\ell}, \quad \ell \in [K]. \quad (19b)$$

Then, for all $\ell, s \in [K]$,

$$s < \ell \implies \Pr_{v \sim \tilde{P}_\ell}[v(s) = 1] \leq r - \eta, \quad s > \ell \implies \Pr_{v \sim \tilde{P}_\ell}[v(s) = 1] \geq r + \eta. \quad (20)$$

The family $(\tilde{P}_\ell)_{\ell \in [K]}$ remains pairwise exactly (ε, δ) -indistinguishable.

Proof. For $z \prec x$, Lemma A.7 gives rank $i - 1$. Since z is not a coordinate of S_x , (C2) and (15) give

$$\Pr_{v \sim P_x} [v(z) = 1] \leq p_{i-1} + \gamma = r - \frac{\Delta_i}{2} + \gamma. \quad (21)$$

For $x \prec z$, the rank is i , and the same homogeneity estimate gives

$$\Pr_{v \sim P_x} [v(z) = 1] \geq p_i - \gamma = r + \frac{\Delta_i}{2} - \gamma. \quad (22)$$

The complete mesh calculation is

$$\begin{aligned} \frac{\Delta_i}{2} - \gamma &\geq \frac{3g_{\text{gap}}}{8M} - \frac{g_{\text{gap}}}{100M} \\ &= \frac{73g_{\text{gap}}}{200M} > \frac{g_{\text{gap}}}{100M} = \frac{a_0}{M} = \eta. \end{aligned} \quad (23)$$

Substituting (23) into (21)–(22) proves (T2), with strictly more margin than required.

Because $p_{i-1} \geq 0$, $p_i \leq 1$, and $\Delta_i > 0$,

$$r = p_{i-1} + \frac{\Delta_i}{2} \geq \frac{\Delta_i}{2} > \eta, \quad r = p_i - \frac{\Delta_i}{2} \leq 1 - \frac{\Delta_i}{2} < 1 - \eta. \quad (24)$$

Thus the two thresholds $r - \eta$ and $r + \eta$ lie in $[0, 1]$.

The maps in (19a)–(19b) give a common deterministic relabeling of every law and coordinate. Hence Proposition A.10 preserves (16), while $j_s \prec j_\ell$ is equivalent to $s < \ell$, proving (20). By (C1),

$$j_1 > j_2 > \cdots > j_K \quad (25)$$

in the natural q -order. Therefore listing J in increasing natural order would reverse both index inequalities and the left/right clauses. Retaining the pullback order, or reversing the list and the clauses simultaneously, preserves the sign. The construction never treats the natural-order reversal as an additional probabilistic or privacy operation.

For $i = 1$, (21) uses p_0 and (22) uses p_1 ; for $i = M$, they use p_{M-1} and p_M . These are genuine homogeneous-list coordinates from the dependency. If one side of x inside J is empty, only that side's universally quantified implication is vacuous. Equation (20), like the source statement, deliberately excludes $s = \ell$.

□

Proposition A.12 (Pairwise private moving-record family). *Under Proposition A.8, set $a_0 = g_{\text{gap}}/100$ and $\eta = a_0/M$. There are a nonempty ordered block $J \subseteq H$, a number $r \in [0, 1]$, and laws $(P_x)_{x \in J}$ on $\{0, 1\}^J$ such that $|J| = |H| - M + 1$, every distinct pair $P_x, P_{x'}$ satisfies both (ε, δ) -privacy inequalities, and, for $z \neq x$,*

$$z \prec x \implies \Pr_{v \sim P_x} \{v(z) = 1\} \leq r - \eta, \quad x \prec z \implies \Pr_{v \sim P_x} \{v(z) = 1\} \geq r + \eta.$$

No condition is imposed on the diagonal coordinate $z = x$; the construction and bounds include $i = 1$ and $i = M$.

Proof. Fix every object quantified by Proposition A.8 conclusion: $n, N, M = 9n, \varepsilon, \delta$, the arbitrary-output exact-private wrapper $\mathcal{W}$, the pullback-ordered homogeneous set H , its list $(p_0, \dots, p_M)$, and any $i \in [M]$ satisfying (A1).

Proposition A.9 removes exactly $M - 1$ fixed extremes from movement and leaves the common ordered block

$$J = \{h_i, \dots, h_{|H|-M+i}\}, \quad |J| = |H| - M + 1. \quad (26)$$

For every $x \in J$, it constructs a threshold-realizable M -record tuple S_x with the moving record at the same ordered position i . Distinct tuples differ in exactly that one record. The proposition explicitly covers the prefix layout $i = 1$ and suffix layout $i = M$, so no interior-index assumption or phantom endpoint point is used.

Lemma A.7 proves that every off-diagonal $z \in J \setminus \{x\}$ sees exactly the two adjacent list ranks: $i - 1$ to the left of x and i to the right. Proposition A.10 applies the wrapper's DP guarantee once to each such database pair and then the same restriction map to J . It therefore gives common-cube arbitrary-output laws $(P_x)_{x \in J}$ satisfying both exact (ε, δ) inequalities, with no composition, group privacy, or parameter degradation.

Finally, Proposition A.11 combines the rank identity with homogeneity and the rise. With the universal choice

$$a_0 = \frac{g_{\text{gap}}}{100}, \quad \eta = \frac{a_0}{M}, \quad r = \frac{p_i + p_{i-1}}{2}, \quad (27)$$

the exact inequality (23) proves the two coordinate margins (T2) for every $x, z \in J, z \neq x$. Reindexing J in the pullback order gives (20), which is the exact lower-index/upper-index family consumed by Proposition A.18. The arbitrary-output coordinate laws, privacy events, and binary-search coordinates are the same objects; no properized or surrogate family is introduced.

This establishes every quantifier and every part of Proposition A.12 conclusion. $\square$

A.4 Binary product contradiction and Ramsey inversion

Work under the temporary contradiction premise

$$\mathcal{R}_n(B, e) < \frac{1}{20} \quad \text{for every } e \in \mathcal{E}_{n,N}. \quad (\text{A1})$$

We prove that there are universal constants

$$a_{\text{th}} > 0, \quad a_\delta > 0, \quad N_{\text{th}} \geq 2 \quad (\text{T1})$$

such that, whenever

$$N \geq N_{\text{th}}, \quad n < a_{\text{th}} \log^* N, \quad 0 < \varepsilon \leq 1, \quad 0 \leq \delta \leq \frac{a_\delta}{n^2 \log(en)}, \quad (\text{T2})$$

every candidate kernel satisfies

$$\max_{e \in \mathcal{E}_{n,N}} \mathcal{R}_n(B, e) \geq \frac{1}{20}. \quad (\text{T3})$$

The endpoint condition inherited from the homogeneous-set argument is

$$\delta \leq \frac{a_{\delta, \text{end}}}{n^2 \log(en)}. \quad (\text{C3})$$

On its nonvacuous Ramsey branch, Proposition A.8 gives the same homogeneous set used below with

$$|H| \geq M + 1, \quad |H| \geq \frac{\log_2^{(M)} N}{\exp(C_R M \log(eM))}. \quad (\text{C4})$$

Put

$$a_0 := \frac{g_{\text{gap}}}{100} > 0, \quad \eta := \frac{a_0}{M}, \quad (\text{C5})$$

and, after the common order-preserving relabeling from Proposition A.12, write

$$J = [K], \quad K = |J| = |H| - M + 1. \quad (\text{C6})$$

Proposition A.12 supplies $a_0 = g_{\text{gap}}/100$, $\eta = a_0/M$, an ordered block $J = [K]$, a number $r \in (\eta, 1 - \eta)$, and pairwise (ε, δ) -indistinguishable laws $(P_x)_{x \in [K]}$ on $\{0, 1\}^{[K]}$ satisfying

$$z < x \implies \mathbb{E}_{V \sim P_x} V(z) \leq r - \eta, \quad z > x \implies \mathbb{E}_{V \sim P_x} V(z) \geq r + \eta. \quad (\text{C7})$$

Lemma A.8 (Product privacy). *If probability laws P, Q satisfy both (ε, δ) -privacy inequalities, then their iid product laws satisfy, for every integer $D \geq 1$ and every event E ,*

$$P^D(E) \leq e^{D\varepsilon} Q^D(E) + D\delta, \quad Q^D(E) \leq e^{D\varepsilon} P^D(E) + D\delta. \quad (\text{C8})$$

Proof. This is the finite-product lemma proved by Alon et al. (2018). Its likelihood-excess decomposition adds one mass- δ defect per product coordinate, giving $D\delta$ rather than an additional geometric factor. $\square$

Lemma A.9 (Hoeffding bound used by the search). *If $Y_1, \dots, Y_D$ are iid in $[0, 1]$ with mean μ , then for $s > 0$,*

$$\Pr \left\{ D^{-1} \sum_{d=1}^D Y_d - \mu \geq s \right\} \leq e^{-2Ds^2}, \quad \Pr \left\{ D^{-1} \sum_{d=1}^D Y_d - \mu \leq -s \right\} \leq e^{-2Ds^2}. \quad (\text{C11})$$

Proof. This is Hoeffding's inequality for bounded independent summands (Hoeffding, 1963). Below it is applied separately to each fixed off-diagonal coordinate on a representative's deterministic ideal path; no independence across coordinates of one output vector is assumed. $\square$

Lemma A.10 (Finite quantitative choice of product size and search depth). *Under the constant $a_0 > 0$ from (C5), let $M \geq 9$,*

$$\eta = \frac{a_0}{M}, \quad L_M := \log(eM), \quad L_0 := \log(9e). \quad (1)$$

There are universal constants $b_T \geq 1$ and $b_D \geq 1$, depending only on the numerical a_0 , such that the finite integers

$$T := \lceil b_T M^2 L_M^2 \rceil, \quad D := \lceil 2\eta^{-2} \log(6T) \rceil \quad (2)$$

satisfy

$$T e^{-2D\eta^2} < \frac{1}{3}, \quad D \leq b_D M^2 L_M, \quad T \log 2 > D + \log 2. \quad (3)$$

Moreover, with $M = 9n$,

$$D \leq 81b_D(1 + \log 9)n^2 \log(en), \quad (4)$$

and

$$T \leq 81(b_T + 1)(1 + \log 9)^2 n^2 \log^2(en). \quad (5)$$

Proof. For an integer $b \geq 1$, define

$$c_D(b) := 2a_0^{-2} \left(4 + \frac{\log(6(b+1))}{L_0} \right) + 1. \quad (6)$$

Because

$$bL_0 \log 2 - c_D(b) - \frac{\log 2}{L_0} \longrightarrow +\infty \quad \text{as } b \longrightarrow \infty, \quad (7)$$

the following set of positive integers is nonempty. Fix b_T to be its least element:

$$b_T L_0 \log 2 > c_D(b_T) + \frac{\log 2}{L_0}. \quad (8)$$

Set

$$b_D := c_D(b_T). \quad (9)$$

These constants are finite and universal because a_0 is the fixed numerical constant $g_{\text{gap}}/100$.

Since $M \geq 9$, one has $L_M \geq L_0 > 1$. From (2),

$$T \leq b_T M^2 L_M^2 + 1 \leq (b_T + 1) M^2 L_M^2. \quad (10)$$

Therefore

$$\begin{aligned} \log(6T) &\leq \log(6(b_T + 1)) + 2 \log M + 2 \log L_M \\ &\leq \left(4 + \frac{\log(6(b_T + 1))}{L_0}\right) L_M, \end{aligned} \quad (11)$$

where $\log M \leq L_M$, $\log L_M \leq L_M$, and $L_M \geq L_0$ were used. Because $\eta^{-2} = a_0^{-2} M^2$, equations (2), (11), and $M^2 L_M \geq 1$ give

$$\begin{aligned} D &\leq 2a_0^{-2} M^2 \log(6T) + 1 \\ &\leq \left[2a_0^{-2} \left(4 + \frac{\log(6(b_T + 1))}{L_0}\right) + 1\right] M^2 L_M \\ &= b_D M^2 L_M. \end{aligned} \quad (12)$$

The defining inequality (8) now gives the strict likelihood/counting separation:

$$\begin{aligned} T \log 2 &\geq b_T M^2 L_M^2 \log 2 \\ &\geq b_T L_0 \log 2 M^2 L_M \\ &> \left(b_D + \frac{\log 2}{L_0}\right) M^2 L_M \\ &\geq D + \log 2. \end{aligned} \quad (13)$$

The last inequality uses (12) and $\log 2 \leq (\log 2/L_0) M^2 L_M$.

For concentration, (2) gives

$$2D\eta^2 \geq 4 \log(6T). \quad (14)$$

Consequently

$$T e^{-2D\eta^2} \leq \frac{T}{(6T)^4} = \frac{1}{1296T^3} < \frac{1}{3}. \quad (15)$$

This proves (3), including strictness.

Finally, $M = 9n$ and $\log(en) \geq 1$ imply

$$L_M = \log(9en) = \log(en) + \log 9 \leq (1 + \log 9) \log(en). \quad (16)$$

Substituting (16) into (10) and (12) proves (4)–(5). Thus no $O(\cdot)$ notation hides a dependence on n , ε , or δ . □

Lemma A.11 (Depth- T search tree with unqueried leaf representatives). *Under the ordered block $J = [K]$ in (C6), let $T \geq 1$ be an integer. If*

$$K > 2^{T+1}, \quad (17)$$

then there is a deterministic full binary search tree of depth T with the following properties:

1. *every node is assigned a nonempty consecutive interval in $[K]$;*

2. every internal node queries one coordinate in its interval and removes that coordinate from both child intervals;
3. the left child lies strictly below the query and the right child lies strictly above it;
4. the 2^T depth- T leaf intervals are nonempty and pairwise disjoint;
5. after choosing one representative x_λ in each leaf interval, the path to that leaf has exactly T query coordinates, all distinct from x_λ .

Proof. For $u \geq 0$, put

$$s_u := 2^{u+1} - 1. \quad (18)$$

An interval of size at least s_u can support a full subtree with u remaining query levels. For $u = 0$, retain any one point as a leaf interval. For $u \geq 1$,

$$s_u = 1 + 2s_{u-1}. \quad (19)$$

Hence an interval of size at least s_u contains a query coordinate leaving at least s_{u-1} coordinates below it and at least s_{u-1} coordinates above it. Assign those two consecutive sides to the children; any extra coordinates may be retained on either side without affecting the lower bound. The query is excluded from both children.

Condition (17) and integrality give

$$K \geq 2^{T+1} + 1 > s_T. \quad (20)$$

Applying the preceding recursion from $u = T$ down to $u = 0$ constructs a full binary tree with exactly 2^T nonempty leaf intervals. Child intervals of different nodes are disjoint, so the leaves are pairwise disjoint. Every root-to-leaf path makes exactly one query at each of the T internal levels. Since each child excludes its parent's query and the representative lies in the final leaf interval, no path query equals its representative. This proves all five claims.

For an observed product sample, route left when the empirical query bit equals one and right when it equals zero. This direction matches (C7): if the target representative is below the query, the query marginal is high; if it is above the query, the query marginal is low. $\square$

Proposition A.13 (Correct adaptive path from fixed-coordinate concentration). *Under (C7), Lemma A.10, and Lemma A.11, draw*

$$V_1, \dots, V_D \stackrel{\text{iid}}{\sim} P_x \quad (21)$$

for a leaf representative x . For every coordinate z , define

$$\widehat{p}(z) := \frac{1}{D} \sum_{d=1}^D V_d(z), \quad \widehat{b}(z) := \mathbf{1}\{\widehat{p}(z) \geq r\}. \quad (22)$$

Let E_x be the event that the routing rule from Lemma A.11 terminates at the leaf represented by x . Then

$$P_x^D(E_x) \geq 1 - Te^{-2D\eta^2} \geq \frac{2}{3}. \quad (23)$$

Proof. Fix x and its designated leaf. Independently of the random vectors, the tree and x determine an ideal path and fixed query coordinates

$$q_1(x), \dots, q_T(x). \quad (24)$$

Lemma A.11 gives $q_t(x) \neq x$ for every t .

If $q_t(x) < x$, then (C7) gives

$$\mu_t := \mathbb{E}_{P_x}[V(q_t(x))] \leq r - \eta.$$

The correct branch is right, which corresponds to $\widehat{b}(q_t(x)) = 0$. Lemma A.9 gives

$$P_x^D[\widehat{b}(q_t(x)) = 1] = P_x^D[\widehat{p}(q_t(x)) \geq r] \leq e^{-2D\eta^2}. \quad (25)$$

If $q_t(x) > x$, then (C7) gives $\mu_t \geq r + \eta$. The correct branch is left, corresponding to bit one, and Lemma A.9 gives

$$P_x^D[\widehat{b}(q_t(x)) = 0] = P_x^D[\widehat{p}(q_t(x)) < r] \leq e^{-2D\eta^2}. \quad (26)$$

Let G_x be the event that all T fixed ideal-path coordinates in (24) have their correct bits. A union bound over these fixed coordinates yields

$$P_x^D(G_x) \geq 1 - Te^{-2D\eta^2}. \quad (27)$$

No coordinate-independence claim is used: for each fixed coordinate, independence is only across the D vectors, and the union bound permits arbitrary dependence among coordinates of one vector.

On G_x , induction over the depth shows that the adaptive router visits the ideal node at every level, hence terminates at the designated leaf. Thus $G_x \subseteq E_x$. Equation (15) proves

$$P_x^D(E_x) \geq P_x^D(G_x) > \frac{2}{3},$$

and the weaker displayed bound (23) follows. The convention in (22) is single-valued even when $\widehat{p}(z) = r$. □

Proposition A.14 (Product-private disjoint-leaf contradiction). *Under the pairwise family (C7), Lemma A.10, Lemma A.11, Proposition A.13, and the local condition*

$$D\delta \leq \frac{1}{6}, \quad (28)$$

one must have

$$K \leq 2^{T+1}. \quad (29)$$

This conclusion is valid uniformly for every $0 < \varepsilon \leq 1$, including $\varepsilon = 1$, and for $\delta = 0$.

Proof. Assume for contradiction that $K > 2^{T+1}$. Lemma A.11 gives exactly 2^T leaf intervals. Choose one representative x_λ per leaf and let E_λ be the event that the deterministic router terminates at that leaf. These events are pairwise disjoint because one product sample follows one binary path and terminates at exactly one leaf.

Proposition A.13 gives

$$P_{x_\lambda}^D(E_\lambda) \geq \frac{2}{3} \quad \text{for every leaf } \lambda. \quad (30)$$

Fix one reference index $y \in [K]$, once and for all. Lemma A.8, applied to P_{x_λ}, P_y , gives

$$P_{x_\lambda}^D(E_\lambda) \leq e^{D\varepsilon} P_y^D(E_\lambda) + D\delta. \quad (31)$$

Solving for the reference-law mass and using (28), (30), and $\varepsilon \leq 1$,

$$\begin{aligned}
P_y^D(E_\lambda) &\geq e^{-D\varepsilon} (P_{x_\lambda}^D(E_\lambda) - D\delta) \\
&\geq e^{-D\varepsilon} \left(\frac{2}{3} - \frac{1}{6} \right) \\
&= \frac{1}{2} e^{-D\varepsilon} \\
&\geq \frac{1}{2} e^{-D}.
\end{aligned} \tag{32}$$

This is the required reference-law leaf-mass inequality. The same single law P_y^D measures every leaf event. At $\varepsilon = 1$ the last bound is exact at the worst endpoint; for $\varepsilon < 1$ it improves. At $\delta = 0$, the additive subtraction in (32) vanishes.

Summing (32) over the 2^T disjoint leaf events gives

$$1 \geq \sum_{\lambda=1}^{2^T} P_y^D(E_\lambda) \geq 2^{T-1} e^{-D}. \tag{33}$$

But (13) is equivalent to

$$(T-1) \log 2 - D > 0,$$

and therefore

$$2^{T-1} e^{-D} > 1, \tag{34}$$

contradicting (33). Thus (29) holds. $\square$

Proposition A.15 (Explicit homogeneous-set upper bound). *Under (A1), Proposition A.8 and Proposition A.12 interfaces, the endpoint delta condition (C3), and the product condition (28), every homogeneous set H satisfies*

$$|H| \leq M - 1 + 2^{T+1} \leq \exp(C_B M^2 \log^2(eM)), \tag{35}$$

where the explicit universal constant

$$C_B := (b_T + 3) \log 2 \tag{36}$$

depends only on the numerical margin.

Proof. The moving-family identity (C6) is exact:

$$K = |H| - M + 1. \tag{37}$$

Proposition A.14 gives $K \leq 2^{T+1}$, proving the first inequality in (35).

Since $b_T \geq 1$, $M \geq 9$, and $L_M > 1$, equation (2) gives $T \geq M^2 \geq M$. Hence

$$M - 1 < 2^{T+1},$$

so

$$|H| \leq 2^{T+2}. \tag{38}$$

Using (10) and $M^2 L_M^2 \geq 1$,

$$\begin{aligned} 2^{T+2} &= \exp((T+2) \log 2) \\ &\leq \exp((b_T + 1) M^2 L_M^2 + 2) \log 2 \\ &\leq \exp((b_T + 3) \log 2 M^2 L_M^2). \end{aligned} \tag{39}$$

This proves (35)–(36). It is the current expected-loss, odd-compatible, $\varepsilon \leq 1$ version of the homogeneous-set upper bound of Alon et al. (2018), with no hidden cardinality term. $\square$

Proposition A.16 (Same-object Ramsey upper/lower comparison). *Under the assumptions of Proposition A.15, suppose also that $N \geq M + 1$ and $\log_2^{(M)} N > 1$ is defined. Then*

$$\log_2^{(M)} N \leq \exp(C_0 M^2 \log^2(eM)), \tag{40}$$

where

$$C_0 := C_B + C_R \tag{41}$$

is universal.

Proof. The Ramsey lower bound (C4) and the upper bound (35) concern the same homogeneous set H , the same sample size M , the same ordered domain, and the same wrapper. Therefore

$$\frac{\log_2^{(M)} N}{\exp(C_R M \log(eM))} \leq |H| \leq \exp(C_B M^2 \log^2(eM)). \tag{42}$$

Multiplying by the positive denominator gives

$$\log_2^{(M)} N \leq \exp(C_B M^2 L_M^2 + C_R M L_M). \tag{43}$$

Since $M \geq 9$ and $L_M > 1$,

$$M L_M \leq M^2 L_M^2. \tag{44}$$

Substituting (44) into (43) proves (40) with the explicit C_0 in (41). No color denominator, additive M , or binary upper-bound term is dropped by prose. $\square$

Lemma A.12 (Finite inversion to a uniform iterated-log lower bound). *Let C_0 be the universal constant in (41). There are universal constants*

$$b_0 := \frac{1}{4}, \quad N_{\text{emp}} \geq 16, \tag{45}$$

such that, for every integer $M \geq 9$ and every $N \geq N_{\text{emp}}$, if

$$M < \log^* N \tag{46}$$

and (40) holds, then

$$M \geq b_0 \log^* N. \tag{47}$$

The definition of N_{emp} below explicitly removes every bounded- M and small- N exception.

Proof. Let $\ell(x) = \log_2 x$ and let $\ell^*(x)$ be its iterated logarithm, exactly the $\log^*$ convention in Section 1. Define

$$E_M := \exp(C_0 M^2 \log^2(eM)). \quad (48)$$

Choose $M_0 \geq 9$ to be the least integer such that, for every integer $M \geq M_0$,

$$\frac{C_0}{\log 2} M^2 \log^2(eM) \leq 2^M \quad (49)$$

and

$$2 + \ell^*(M) \leq M. \quad (50)$$

Such an integer exists because $2^M / (M^2 \log^2(eM))$ tends to infinity and $\ell^*(M) = o(M)$. Equations (49)–(50), rather than an unquantified “large enough” phrase, are the defining threshold conditions.

For every $M \geq M_0$, (49) gives

$$\ell(E_M) = \frac{C_0}{\log 2} M^2 \log^2(eM) \leq 2^M. \quad (51)$$

By monotonicity of ℓ^* ,

$$\ell^*(E_M) = 1 + \ell^*(\ell(E_M)) \leq 1 + \ell^*(2^M) = 2 + \ell^*(M) \leq M. \quad (52)$$

The remaining integers $9 \leq M < M_0$ form a finite set. Define

$$R_{\text{fin}} := \max_{9 \leq M < M_0} (M + \ell^*(E_M)), \quad (53)$$

with maximum zero if the set is empty. Choose $N_{\text{emp}} \geq 16$ so that

$$\ell^*(N_{\text{emp}}) > R_{\text{fin}}. \quad (54)$$

This is a finite universal choice because ℓ^* is unbounded.

Now assume (46) and (40). Condition (46) implies $\ell^{(M)}(N) > 1$, and the defining recursion of ℓ^* gives the exact identity

$$\ell^*(N) = M + \ell^*(\ell^{(M)}N). \quad (55)$$

By (40), monotonicity, and (48),

$$\ell^*(N) \leq M + \ell^*(E_M). \quad (56)$$

If $M < M_0$, then (53)–(56) give

$$\ell^*(N) \leq R_{\text{fin}},$$

whereas $N \geq N_{\text{emp}}$ and (54) give $\ell^*(N) \geq \ell^*(N_{\text{emp}}) > R_{\text{fin}}$, a contradiction. Thus $M \geq M_0$. Equation (52) then turns (56) into

$$\ell^*(N) \leq 2M. \quad (57)$$

Hence

$$M \geq \frac{1}{2} \ell^*(N) > \frac{1}{4} \ell^*(N) = b_0 \log^* N, \quad (58)$$

which proves the weaker interface (47) with strict slack. The threshold N_{emp} is the only small- N exclusion, and it is universal. $\square$

Proposition A.17 (One-arm threshold obstruction). *Under the primitive finite-experiment setting and the conclusions of Propositions A.3, A.8, and A.12, define*

$$C_9 := 81(1 + \log 9), \quad (59)$$

$$a_\delta := \min \left\{ a_{\delta, \text{end}}, \frac{1}{6b_D C_9} \right\} > 0, \quad (60)$$

$$a_{\text{th}} := \frac{b_0}{9} = \frac{1}{36}, \quad N_{\text{th}} := N_{\text{emp}}. \quad (61)$$

Then (T2) implies (T3) for every arbitrary full-cube (ε, δ) -DP kernel B .

Proof. All constants in (59)–(61) are universal. The constant $a_{\delta, \text{end}}$ comes from Proposition A.8; b_D comes from Lemma A.10; and b_0, N_{emp} come from Lemma A.12.

Fix parameters satisfying (T2) and a candidate kernel B . Suppose (A1) holds. Since $a_\delta \leq a_{\delta, \text{end}}$,

$$\delta \leq \frac{a_\delta}{n^2 \log(en)} \leq \frac{a_{\delta, \text{end}}}{n^2 \log(en)}, \quad (62)$$

so the endpoint-gap and moving-family dependencies apply once their finite Ramsey entry conditions are checked.

The sample-size premise and $M = 9n$ give the strict inequality

$$M = 9n < 9a_{\text{th}} \log^* N = b_0 \log^* N = \frac{1}{4} \log^* N < \log^* N. \quad (63)$$

Thus $\log_2^{(M)} N > 1$ is defined. Also $\log^* N \leq N$ for $N \geq 2$, so (63) gives $M < N$; because both are integers,

$$N \geq M + 1. \quad (64)$$

Equations (63)–(64) discharge every nonvacuous finite-Ramsey entry condition in Proposition A.8.

For the product defect, (4), (59), (60), and (T2) give

$$\begin{aligned} D\delta &\leq b_D M^2 \log(eM) \frac{a_\delta}{n^2 \log(en)} \\ &\leq b_D C_9 a_\delta \\ &\leq \frac{1}{6}. \end{aligned} \quad (65)$$

The middle inequality is the exact comparison

$$M^2 \log(eM) = 81n^2 \log(9en) \leq C_9 n^2 \log(en), \quad (66)$$

already proved through (16). At $\delta = 0$, the left-hand side of (65) is exactly zero.

The dependencies now produce $H, J, r, (P_x)$ under (A1). Lemma A.10, Proposition A.14, and Proposition A.15 give the homogeneous-set upper bound. Proposition A.16 gives (40). Since $N \geq N_{\text{th}} = N_{\text{emp}}$ and (63) gives $M < \log^* N$, Lemma A.12 gives

$$M \geq b_0 \log^* N. \quad (67)$$

But (63) gives the strict opposite inequality

$$M < b_0 \log^* N. \quad (68)$$

This contradiction discharges (A1). Therefore there is an experiment $e \in \mathcal{E}_{n,N}$ such that

$$\mathcal{R}_n(B, e) \geq \frac{1}{20},$$

which is (T3).

The proof used $\varepsilon \leq 1$ only in the endpoint-uniform gap and in the explicit likelihood comparison $e^{-D\varepsilon} \geq e^{-D}$. Thus $\varepsilon = 1$ is included with no limiting argument. Every output remains an arbitrary bit vector; only coordinate averages and events are used. The entire argument is for one fixed candidate kernel and never introduces a prior. $\square$

Proposition A.18 (One-arm threshold obstruction). *There are universal constants $a_{\text{th}}, a_\delta > 0$ and an integer $N_{\text{th}} \geq 2$ such that, whenever*

$$N \geq N_{\text{th}}, \quad n < a_{\text{th}} \log^* N, \quad 0 < \varepsilon \leq 1, \quad 0 \leq \delta \leq \frac{a_\delta}{n^2 \log(en)},$$

every full-cube arbitrary-output (ε, δ) -private threshold-domain kernel B satisfies

$$\max_{e \in \mathcal{E}_{n,N}} \mathcal{R}_n(B, e) \geq \frac{1}{20}.$$

The conclusion is learner by learner and does not assume or produce a prior.

Proof. Fix a candidate arbitrary-output (ε, δ) -DP n -sample threshold kernel and parameters satisfying (T2). Proposition A.3 restricts the output to the full cube, preserves privacy and every finite-experiment risk, and converts the local negation of (T3) into the exact expected empirical-loss premise (A1) for the $M = 9n$ wrapper.

The explicit choice (60) first discharges the endpoint delta condition of Proposition A.8. The strict sample relation (63) discharges the finite Ramsey entry conditions, so Proposition A.8 produces a homogeneous set with the exact lower bound (C4), and Proposition A.12 produces the same-target family (C7) on a block of exact size $|H| - M + 1$.

Lemma A.10 chooses finite D, T and proves, before any event construction,

$$Te^{-2D\eta^2} < \frac{1}{3}, \quad D \leq b_D M^2 \log(eM), \quad T \log 2 > D + \log 2.$$

Lemma A.11 gives exactly 2^T nonempty disjoint leaves whenever the block is too large and leaves one unqueried representative in every leaf. Proposition A.13 applies Hoeffding only to the T fixed coordinates on each representative's ideal path; arbitrary within-vector coordinate dependence is harmless. Thus the correct leaf has representative-law mass at least $2/3$.

Lemma A.8, fully instantiated in Proposition A.14, gives the finite product certificate $(D\varepsilon, D\delta)$. Equation (65) caps the adversarial-sign additive defect at $1/6$, and the worst endpoint $\varepsilon = 1$ leaves every leaf mass at least $(1/2)e^{-D}$ under one fixed reference law. Disjointness and the strict inequality $2^{T-1}e^{-D} > 1$ force the exact upper bound

$$|H| \leq M - 1 + 2^{T+1}.$$

Proposition A.15 turns this into an explicit exponential bound. Proposition A.16 compares it with the Ramsey lower bound on the identical set H , retaining the color denominator through C_R . Lemma A.12 handles both unbounded and finitely many bounded M cases, yielding $M \geq b_0 \log^* N$ for $N \geq N_{\text{emp}}$. Proposition A.17 then uses $M = 9n$ and the strict premise $n < a_{\text{th}} \log^* N$ to obtain the contradiction.

Therefore the conclusion holds with the universal constants in (60)–(61). The result is the algorithm-wise finite-game lower value and its threshold constants. No minimax theorem, learner-independent prior, hidden-arm construction, or hard-prior assumption appears in this step. $\square$

A.5 Finite minimax and hard prior

Let $a_{\text{th}}, a_\delta > 0$ and $N_{\text{th}} \geq 2$ be the universal constants of Proposition A.18. Throughout this subsection assume

$$N \geq N_{\text{th}}, \quad n < a_{\text{th}} \log^* N, \quad 0 < \varepsilon \leq 1, \quad 0 \leq \delta \leq \frac{a_\delta}{n^2 \log(en)}. \quad (\text{T1})$$

Theorem A.2 (Finite-dimensional minimax). *If X and Y are nonempty compact convex subsets of finite-dimensional real vector spaces and $F : X \times Y \rightarrow \mathbb{R}$ is continuous and affine in each argument, then*

$$\min_{x \in X} \max_{y \in Y} F(x, y) = \max_{y \in Y} \min_{x \in X} F(x, y), \quad (\text{C2})$$

and both outer extrema are attained.

Proof. This is the finite-dimensional minimax theorem (von Neumann, 1928). The stochastic-kernel polytope and finite probability simplex below meet its hypotheses, and their payoff is bilinear. $\square$

Proposition A.19 (Compact convex polytope of finite private kernels). *Under the primitive finite-game definitions, fix $n \geq 1$, $N \geq 2$, $\varepsilon > 0$, and $\delta \geq 0$. Define the labeled alphabet, ordered input space, unrestricted binary output cube, and experiment universe by*

$$\mathbf{Z}_N := [N] \times \{0, 1\}, \quad \mathbf{S}_{n,N} := \mathbf{Z}_N^n, \quad (1)$$

$$\mathbf{V}_N := \{0, 1\}^{[N]}, \quad \mathbf{E}_{n,N} := \mathcal{E}_{n,N} = [N] \times [N]^{9n}. \quad (2)$$

Let $s \sim s'$ mean that the ordered samples $s, s' \in \mathbf{S}_{n,N}$ differ in at most one coordinate. Let $\mathcal{K}_{n,N}^{\varepsilon,\delta}$ be the set of arrays

$$K = (K(v \mid s))_{s \in \mathbf{S}_{n,N}, v \in \mathbf{V}_N} \quad (3)$$

satisfying, for every s ,

$$K(v \mid s) \geq 0 \quad \text{for all } v, \quad \sum_{v \in \mathbf{V}_N} K(v \mid s) = 1, \quad (4)$$

and, for every replacement-adjacent pair $s \sim s'$ and every event $A \subseteq \mathbf{V}_N$, both eventwise DP inequalities

$$\sum_{v \in A} K(v \mid s) \leq e^\varepsilon \sum_{v \in A} K(v \mid s') + \delta, \quad (5a)$$

$$\sum_{v \in A} K(v \mid s') \leq e^\varepsilon \sum_{v \in A} K(v \mid s) + \delta. \quad (5b)$$

Then $\mathcal{K}_{n,N}^{\varepsilon,\delta}$ is a nonempty compact convex polytope in a finite-dimensional Euclidean space.

Proof. All four sets in (1)–(2) are finite, with

$$|\mathbf{Z}_N| = 2N, \quad |\mathbf{S}_{n,N}| = (2N)^n, \quad |\mathbf{V}_N| = 2^N, \quad |\mathbf{E}_{n,N}| = N^{9n+1}. \quad (6)$$

Thus a kernel is a point in

$$\mathbb{R}^d, \quad d = (2N)^n 2^N < \infty. \quad (7)$$

Equations (4), (5a), and (5b) list every stochastic-row and every eventwise privacy requirement explicitly. There are finitely many such constraints: the adjacent-pair set is finite and the finite output cube has only finitely many events.

For nonemptiness, let $v_0 \in \mathbf{V}_N$ be the all-zero vector and define the input-independent deterministic kernel

$$K_0(v \mid s) := \mathbf{1}\{v = v_0\}. \quad (8)$$

It satisfies (4). For any event A , either both event probabilities in (5a)–(5b) equal one or both equal zero. In the first case $1 \leq e^\varepsilon + \delta$; in the second case $0 \leq \delta$. Hence K_0 is (ε, δ) -DP. This also verifies nonemptiness at the boundary $\delta = 0$, including $\varepsilon = 1$.

The row-sum and nonnegativity conditions imply

$$0 \leq K(v \mid s) \leq 1 \quad \text{for every } (s, v), \quad (9)$$

so the set is bounded. Every row-sum condition is a closed affine hyperplane, every nonnegativity condition is a closed halfspace, and each of (5a)–(5b) is a closed affine halfspace. Their finite intersection is closed. By the finite-dimensional Heine-Borel theorem, the set is compact. Because it is defined by finitely many affine equalities and inequalities, it is a polytope.

Finally, take $K_1, K_2 \in \mathcal{K}_{n,N}^{\varepsilon,\delta}$ and $\lambda \in [0, 1]$, and put

$$K_\lambda := \lambda K_1 + (1 - \lambda) K_2. \quad (10)$$

Nonnegativity and row sums are preserved. For every $s \sim s'$ and A ,

$$\begin{aligned} K_\lambda(A \mid s) &= \lambda K_1(A \mid s) + (1 - \lambda) K_2(A \mid s) \\ &\leq \lambda(e^\varepsilon K_1(A \mid s') + \delta) + (1 - \lambda)(e^\varepsilon K_2(A \mid s') + \delta) \\ &= e^\varepsilon K_\lambda(A \mid s') + \delta, \end{aligned} \quad (11)$$

and the same calculation with s, s' exchanged gives the reverse inequality. Thus K_λ belongs to the set, proving convexity. Notice that the two additive terms combine to $\lambda\delta + (1 - \lambda)\delta = \delta$, so convexification causes no privacy loss. $\square$

Proposition A.20 (Affine finite risk and exact arbitrary-output restriction). *Under the primitive experiment definitions and the arbitrary-output finite-restriction interface, for every $e = (t, U) \in \mathbf{E}_{n,N}$, the risk $r_e(K) = \mathcal{R}_n(K, e)$ is a continuous affine function of $K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}$. Moreover, pushing an arbitrary-output (ε, δ) -DP kernel through the $[N]$ -coordinate restriction preserves both privacy parameters and every r_e exactly, while every finite-cube kernel is itself an allowed arbitrary-output kernel. Consequently, for every prior $\Pi \in \Delta(\mathbf{E}_{n,N})$, the arbitrary-output and finite-cube prior-average infima are equal.*

Proof. Write

$$U = (u_1, \dots, u_{9n}).$$

The experiment distribution, including all multiplicities in U , is

$$Q_e(q, y) = \frac{1}{9n} \sum_{a=1}^{9n} \mathbf{1}\{q = u_a, y = \tau_t(u_a)\}, \quad (q, y) \in \mathbf{Z}_N. \quad (12)$$

For $v \in \mathbf{V}_N$, define its exact experiment loss

$$\ell_e(v) := L_{Q_e}(v) = \sum_{(q,y) \in \mathbf{Z}_N} Q_e(q,y) \mathbf{1}\{v(q) \neq y\} = \frac{1}{9n} \sum_{a=1}^{9n} \mathbf{1}\{v(u_a) \neq \tau_t(u_a)\}. \quad (13)$$

For an ordered sample $s = (z_1, \dots, z_n) \in \mathbf{S}_{n,N}$, put

$$Q_e^n(s) := \prod_{i=1}^n Q_e(z_i). \quad (14)$$

Then the kernel risk is the finite sum

$$\begin{aligned} r_e(K) &:= \sum_{s \in \mathbf{S}_{n,N}} Q_e^n(s) \sum_{v \in \mathbf{V}_N} K(v | s) \ell_e(v) \\ &= \mathbb{E}_{S \sim Q_e^n, V \sim K(S)} [L_{Q_e}(V)] = \mathcal{R}_n(K, e). \end{aligned} \quad (15)$$

All coefficients $Q_e^n(s) \ell_e(v)$ are fixed once e is fixed, so (15) is linear, hence continuous and affine, in the coordinates of K . Also

$$0 \leq r_e(K) \leq 1. \quad (16)$$

We now apply only the finite restriction. Let an arbitrary-output kernel be

$$\tilde{B} : \mathbf{S}_{n,N} \rightsquigarrow (\mathbf{Y}, \mathcal{A}), \quad (17)$$

where its output has measurable binary evaluations on $[N]$. Let

$$\rho : \mathbf{Y} \longrightarrow \mathbf{V}_N, \quad \rho(y)(q) := \text{the prediction of } y \text{ at } q, \quad (18)$$

be the restriction map. It is measurable directly from the setting convention: for every $v \in \mathbf{V}_N$,

$$\rho^{-1}(\{v\}) = \bigcap_{q \in [N]} \{y \in \mathbf{Y} : \rho(y)(q) = v(q)\}, \quad (18a)$$

a finite intersection of measurable coordinate events, and every subset of the finite cube is a finite union of such singletons. For clarity, eventwise privacy of $\tilde{B}$ means that for every $s \sim s'$ and every measurable $F \in \mathcal{A}$, both

$$\tilde{B}(F | s) \leq e^\varepsilon \tilde{B}(F | s') + \delta, \quad \tilde{B}(F | s') \leq e^\varepsilon \tilde{B}(F | s) + \delta. \quad (19)$$

Define the pushforward finite kernel by

$$K_\rho(A | s) := \tilde{B}(\rho^{-1}(A) | s), \quad A \subseteq \mathbf{V}_N. \quad (20)$$

Because ρ is measurable, substituting $F = \rho^{-1}(A)$ into both inequalities in (19) proves exactly (5a)–(5b) for K_ρ . Thus

$$K_\rho \in \mathcal{K}_{n,N}^{\varepsilon,\delta} \quad (21)$$

with the same ε, δ ; there is no composition, group-privacy, or parameter relaxation.

Every Q_e is supported on $[N]$, and its loss depends on an output only through the vector (18). Therefore

$$\begin{aligned} \mathcal{R}_n(\tilde{B}, e) &= \sum_s Q_e^n(s) \int_{\mathbf{Y}} \ell_e(\rho(y)) \tilde{B}(dy | s) \\ &= \sum_s Q_e^n(s) \sum_{v \in \mathbf{V}_N} K_\rho(v | s) \ell_e(v) \\ &= r_e(K_\rho). \end{aligned} \quad (22)$$

This is exact even when U contains repeated entries and even when the arbitrary hypothesis oscillates completely away from the threshold target.

Conversely, every $K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}$ is already an arbitrary-output kernel by taking the output space to be the full discrete cube $\mathbf{Y} = \mathbf{V}_N$ and ρ to be the identity. The cube contains all binary functions on $[N]$, not only thresholds. Hence the pushforward map is onto the finite competitor class relevant to the game.

For a prior Π , define

$$\Phi(K, \Pi) := \sum_{e \in \mathbf{E}_{n,N}} \Pi(e) r_e(K). \quad (23)$$

Equation (22) maps every arbitrary-output competitor to a finite competitor with exactly the same payoff, while the converse realization includes every finite competitor among arbitrary outputs. Therefore

$$\inf_{\tilde{B}: (\varepsilon, \delta)\text{-DP}} \mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(\tilde{B}, e)] = \min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \Phi(K, \Pi). \quad (24)$$

The right side is a minimum, not merely an infimum, because Proposition A.19 gives compactness and (23) is continuous. Thus arbitrary outputs are handled with exactly the finite restriction and with zero privacy, risk, support, or discretization residual. $\square$

Proposition A.21 (Correctly oriented finite minimax with attainment). *Under Propositions A.19 and A.20, let $\Delta(\mathbf{E}_{n,N})$ be the probability simplex on the exact finite experiment set. Then*

$$\boxed{\min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \max_{e \in \mathbf{E}_{n,N}} r_e(K) = \max_{\Pi \in \Delta(\mathbf{E}_{n,N})} \min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \mathbb{E}_{e \sim \Pi}[r_e(K)]} \quad (25)$$

with the learner minimization on the left and the adversarial prior maximization on the right. The left minimum is attained, the inner minimum is attained for every prior, and the right maximum is attained by at least one prior.

Proof. The finite simplex is

$$\Delta(\mathbf{E}_{n,N}) = \left\{ \Pi \in \mathbb{R}^{|\mathbf{E}_{n,N}|} : \Pi(e) \geq 0, \sum_e \Pi(e) = 1 \right\}. \quad (26)$$

It is nonempty, compact, and convex. Proposition A.19 gives the same three properties for $\mathcal{K}_{n,N}^{\varepsilon,\delta}$. By (23),

$$\Phi(K, \Pi) = \sum_e \Pi(e) r_e(K) \quad (27)$$

is bilinear and continuous: it is linear in Π , and every r_e is linear in K by (15). Thus all hypotheses of Theorem A.2 are discharged, giving

$$\min_K \max_{\Pi} \Phi(K, \Pi) = \max_{\Pi} \min_K \Phi(K, \Pi). \quad (28)$$

For each fixed K , linearity over the simplex and finiteness of the experiment set give

$$\max_{\Pi \in \Delta(\mathbf{E}_{n,N})} \Phi(K, \Pi) = \max_{e \in \mathbf{E}_{n,N}} r_e(K), \quad (29)$$

and the maximum is attained by a point mass on any maximizing experiment. Substituting (29) into (28) proves exactly (25), including its orientation.

For completeness, all required attainment statements can also be seen directly. The function

$$K \mapsto \max_e r_e(K) \quad (30)$$

is the maximum of finitely many continuous functions, so it is continuous and attains its minimum on the compact kernel polytope. For each fixed Π , (27) is continuous in K , so the inner minimum is attained. Finally, define

$$g(\Pi) := \min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \Phi(K, \Pi). \quad (31)$$

Using (16), for any priors Π, Π' ,

$$\sup_K |\Phi(K, \Pi) - \Phi(K, \Pi')| \leq \sum_{e \in \mathcal{E}_{n,N}} |\Pi(e) - \Pi'(e)|. \quad (32)$$

Taking minima in K on both sides in the standard two directions yields

$$|g(\Pi) - g(\Pi')| \leq \|\Pi - \Pi'\|_1. \quad (33)$$

Thus g is continuous and attains its maximum on the compact simplex. No measurable-selection or limiting argument is involved. $\square$

Proposition A.22 (Attained learner-independent hard prior). *Under the admissibility conditions (T1), Proposition A.17, Proposition A.20, and Proposition A.21, there exists an attaining prior*

$$\Pi_{n,N,\varepsilon,\delta} \in \Delta(\mathcal{E}_{n,N}) \quad (34)$$

such that

$$\inf_{B: (\varepsilon,\delta)\text{-DP}} \mathbb{E}_{e \sim \Pi_{n,N,\varepsilon,\delta}} [\mathcal{R}_n(B, e)] \geq \frac{1}{20}. \quad (35)$$

The prior depends only on the fixed game parameters $n, N, \varepsilon, \delta$, is chosen after the learner-by-learner obstruction is proved, and assigns probability only to the exact experiments $e = (t, U) \in [N] \times [N]^{9n}$.

Proof. Proposition A.17 gives, for every $K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}$,

$$\max_{e \in \mathcal{E}_{n,N}} r_e(K) \geq \frac{1}{20}. \quad (36)$$

Taking the minimum over the entire compact kernel set preserves the same constant:

$$\min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \max_{e \in \mathcal{E}_{n,N}} r_e(K) \geq \frac{1}{20}. \quad (37)$$

Proposition A.21 now gives an attaining prior $\Pi_{n,N,\varepsilon,\delta}$ for which

$$\begin{aligned} \min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \mathbb{E}_{e \sim \Pi_{n,N,\varepsilon,\delta}} [r_e(K)] &= \min_{K \in \mathcal{K}_{n,N}^{\varepsilon,\delta}} \max_{e \in \mathcal{E}_{n,N}} r_e(K) \\ &\geq \frac{1}{20}. \end{aligned} \quad (38)$$

There is no degradation of the threshold: the exact algorithm-wise value in (36) passes through equality (25).

Applying the exact arbitrary-output equality (24) to this same prior turns (38) into

$$\inf_{B: (\varepsilon, \delta)\text{-DP}} \mathbb{E}_{e \sim \Pi_{n, N, \varepsilon, \delta}} [\mathcal{R}_n(B, e)] = \min_{K \in \mathcal{K}_{n, N}^{\varepsilon, \delta}} \mathbb{E}_{e \sim \Pi_{n, N, \varepsilon, \delta}} [r_e(K)] \geq \frac{1}{20}, \quad (39)$$

which is (HP) with $\eta_{\text{th}} = 1/20$ exactly.

The prior in (34) is a probability vector indexed by precisely $\mathcal{E}_{n, N} = [N] \times [N]^{9n}$. Hence it has no mass on an auxiliary discretization, limiting experiment, changed database size, or different distribution class. As usual for notation $\Delta(\mathcal{E}_{n, N})$, its positive-mass support may be a subset of that exact finite universe; (HP) does not require every experiment to receive positive mass.

Finally, the construction is noncircular. Proposition A.17 proves (36) separately for every candidate kernel without mentioning a prior. Only after the entire risk table $(r_e(K))_{K, e}$ and the full learner strategy set are fixed does minimax select one maximizing prior. The selected probability vector may depend on $n, N, \varepsilon, \delta$, but it does not depend on a later candidate learner, the hidden-arm learner, the PAC learner A , or any randomness generated by those learners. This is the first construction of $\Pi_{n, N, \varepsilon, \delta}$. □

Proposition A.23 (Learner-independent hard prior). *Under the scalar conditions (T1), there is an attaining prior*

$$\Pi_{n, N, \varepsilon, \delta} \in \Delta(\mathcal{E}_{n, N})$$

depending only on $(n, N, \varepsilon, \delta)$ such that

$$\inf_{B: (\varepsilon, \delta)\text{-DP}} \mathbb{E}_{e \sim \Pi_{n, N, \varepsilon, \delta}} [\mathcal{R}_n(B, e)] \geq \frac{1}{20}. \quad (\text{HP})$$

The infimum ranges over all coordinate-measurable arbitrary-output threshold-domain kernels; no properness or computational restriction is imposed.

Proof. Fix parameters satisfying (T1). Proposition A.19 defines the learner's finite strategy set using every stochastic-row condition and both eventwise DP inequalities on every ordered replacement-adjacent input pair and every event in the full output cube. The constant kernel proves nonemptiness; finite affine constraints give closedness; stochasticity gives boundedness; Heine-Borel gives compactness; and direct convex combination proves convexity without changing ε or δ .

Proposition A.20 writes each risk as the explicit finite affine sum (15), including the multiplicities in U . It then applies precisely the coordinate restriction to push every arbitrary-output DP kernel into the finite cube, proves both privacy inequalities through event preimages, and proves equality of every experiment risk. Because every cube vector is itself an arbitrary binary hypothesis on $[N]$, the restriction is onto the relevant finite competitor class and (24) gives equality of the arbitrary-output and finite-game infima. No threshold properness, representation restriction, approximation, or discretization is introduced.

Proposition A.21 applies the fully restated Theorem A.2 to the compact convex DP polytope and the exact finite experiment simplex. Its payoff is continuous bilinear, and maximizing over the experiment simplex is exactly maximizing over experiments. Therefore the orientation is

$$\min_K \max_e r_e(K) = \max_{\Pi} \min_K \mathbb{E}_{e \sim \Pi} r_e(K),$$

not the reverse. The learner-side minimum, each fixed-prior inner minimum, and the adversary-side maximum are attained.

Proposition A.17 lower-bounds the left side by exactly $1/20$ before any prior exists. Proposition A.22 chooses an attaining prior only afterward, transfers the same value through minimax

without loss, and then invokes the exact arbitrary-output equality. This proves (HP) for a learner-independent $\Pi_{n,N,\varepsilon,\delta} \in \Delta(\mathcal{E}_{n,N})$, supported on no experiments outside the exact finite universe. All privacy parameters, risks, experiment distributions, and the threshold $1/20$ are unchanged.

Thus the conclusion of Proposition A.23 is proved, and the prior is available for Proposition A.27 and later arguments. □

A.6 Hidden-arm kernel and exact privacy

Put

$$n = \left\lceil \frac{2m}{k} \right\rceil + 12$$

and fix the prior produced by Proposition A.23,

$$\Pi = \Pi_{n,N,\varepsilon_0,\delta_m} \in \Delta(\mathcal{E}_{n,N}). \quad (\text{T1})$$

Only the fact that Π is a fixed learner-independent probability law is used to construct the kernel; its hard-value inequality is not used as privacy evidence. The primitive privacy interface for the learner is, for every ordered replacement-adjacent pair x, x' and every measurable output event F ,

$$A(F \mid x) \leq e^{\varepsilon_0} A(F \mid x') + \delta_m, \quad A(F \mid x') \leq e^{\varepsilon_0} A(F \mid x) + \delta_m. \quad (\text{C1})$$

The kernel constructed below will satisfy, for every replacement-adjacent s, s' and output event G ,

$$B_\Pi(G \mid s) \leq e^{\varepsilon_0} B_\Pi(G \mid s') + \delta_m, \quad B_\Pi(G \mid s') \leq e^{\varepsilon_0} B_\Pi(G \mid s) + \delta_m. \quad (\text{T2})$$

Lemma A.13 (Iid completion of the hidden-arm latent experiments). *Under Assumption 1 and the prior (T1), let J be uniform on $[k]$. The latent experiment draws used outside arm J can be completed by one unused coordinate so that*

$$\mathbf{E} = (E_1, \dots, E_k) \sim \Pi^k \quad \text{and} \quad \mathbf{E} \text{ is independent of } J. \quad (1)$$

Moreover, in the prior-average risk experiment, if an outer experiment $E_* \sim \Pi$ generates the real input and one sets $E_J = E_*$ while drawing $E_j \sim \Pi$ independently for $j \neq J$, then the same conclusion (1) holds and the nonhidden draws have exactly the law used by B_Π . This remains valid at $k = 1$.

Proof. For the kernel's internal implementation, draw

$$J \sim \text{Unif}([k]), \quad (E_1, \dots, E_k) \sim \Pi^k \quad (2)$$

independently of one another and independently of the real input. The construction below uses only E_j with $j \neq J$; hence the unused coordinate E_J can be included without changing the procedure stated in Section 1, which draws only the nonhidden coordinates. Equation (1) is immediate from (2), but we record the exact joint mass because it is the exchangeability interface used later. For every $a \in [k]$ and every $\mathbf{e} = (e_1, \dots, e_k) \in \mathcal{E}_{n,N}^k$,

$$\Pr\{J = a, \mathbf{E} = \mathbf{e}\} = \frac{1}{k} \prod_{j=1}^k \Pi(e_j). \quad (3)$$

The right side factors into $\Pr J = a$ times the product-law mass, proving independence even when some prior masses vanish.

For the outer prior-average completion, draw J uniformly, draw $E_* \sim \Pi$ independently, set $E_J = E_*$, and, conditional on J , draw the remaining coordinates independently from Π . Then again

$$\Pr\{J = a, E_1 = e_1, \dots, E_k = e_k\} = \frac{1}{k} \Pi(e_a) \prod_{j \neq a} \Pi(e_j) = \frac{1}{k} \prod_{j=1}^k \Pi(e_j). \quad (4)$$

Thus the completed vector is iid and independent of J . If the real sample is then drawn from $Q_{E_*}^n$, its conditional law given $(J, \mathbf{E})$ is exactly $Q_{E_J}^n$.

This coupling does not alter B_Π : the hidden coordinate is unused by the kernel, and all used nonhidden coordinates have the same independent Π law as in (2). At $k = 1$, the product over $j \neq J$ is empty, $J = 1$ deterministically, and the completed vector is simply $(E_*) \sim \Pi$; independence from a constant variable and the singleton product law are exact. $\square$

Proposition A.24 (Measurable arbitrary-output hidden-arm kernel). *Under Assumptions 1, 2, and 3, the prior (T1), and Lemma A.13, the following procedure defines a Markov kernel*

$$B_\Pi : Z_N^n \rightsquigarrow V_N, \quad Z_N := [N] \times \{0, 1\}, \quad V_N := \{0, 1\}^{[N]}, \quad (5)$$

on every ordered labeled input. Its output is the measurable restriction of an arbitrary hypothesis produced by A ; no threshold shape, properness, representation, or computational condition is imposed.

Proof. For each arm define the record transport and hypothesis restriction

$$T_j(q, y) := (\phi_j(q), y) \in Z_X, \quad \rho_j(h)(q) := h(\phi_j(q)), \quad q \in [N]. \quad (6)$$

The maps T_j are legal because Assumption 1 gives $\phi_j : [N] \rightarrow X$. For restriction measurability, fix j and $v \in V_N$. By the coordinate-measurability clause in Assumption 2,

$$\rho_j^{-1}(\{v\}) = \bigcap_{q \in [N]} \{h : h(\phi_j(q)) = v(q)\} \quad (7)$$

is measurable. Every event $G \subseteq V_N$ is a finite union of singletons, so $\rho_j^{-1}(G)$ is measurable as well.

On a real input

$$s = (z_1, \dots, z_n) \in Z_N^n, \quad (8)$$

draw the data-independent variables in (2), draw

$$I_1, \dots, I_m \stackrel{\text{iid}}{\sim} \text{Unif}([k]) \quad (9)$$

independently, and put

$$H := \{r \in [m] : I_r = J\} = \{h_1 < \dots < h_R\}, \quad R := |H|. \quad (10)$$

For every nonhidden position r with $I_r = j \neq J$, independently draw

$$W_r \sim Q_{E_j}. \quad (11)$$

All variables in (2), (9), and (11) are independent of s . If $R > n$, output the fixed vector

$$g_0(q) := 0, \quad q \in [N]. \quad (12)$$

If $R \leq n$, define the ordered input to A by

$$F_\omega(s)_r := \begin{cases} T_J(z_a), & r = h_a \text{ for some } a \in [R], \\ T_{I_r}(W_r), & I_r \neq J, \end{cases} \quad r \in [m], \quad (13)$$

where

$$\omega = (J, \mathbf{E}, (I_r)_{r=1}^m, (W_r)_{I_r \neq J}) \quad (14)$$

denotes the internal transcript. Run $h \sim A(F_\omega(s))$ and output

$$g = \rho_J(h) \in \mathbb{V}_N. \quad (15)$$

Conditional on every internal transcript, (7) makes (15) a measurable postprocessing of A ; on overflow the output is the measurable constant (12). The input space in (5) and the output cube are finite, and averaging these conditional output laws over the finite data-independent transcript law therefore defines a Markov kernel.

The unused draw E_J in (2) does not enter (11) or (13). Consequently, this full-vector implementation has exactly the same kernel law as the procedure in Section 1, which draws only E_j for $j \neq J$. The output cube contains every binary function on $[N]$, so (15) does not properize the learner or force a threshold. The construction is defined on all labeled samples (8), including nonrealizable ones, as required by differential privacy. $\square$

Lemma A.14 (One-use transcript locality under ordered replacement). *Under Proposition A.24, fix any internal transcript ω and any replacement-adjacent inputs $s, s' \in \mathbb{Z}_N^n$. If $R > n$, both conditional kernels output the same constant g_0 . If $R \leq n$, then $F_\omega(s)$ and $F_\omega(s')$ are equal or replacement-adjacent ordered m -samples. More precisely, when s, s' differ at the unique coordinate ℓ , the constructed learner inputs are equal if $\ell > R$ and differ only at position h_ℓ if $\ell \leq R$. This covers $R = 0$, $R = n$, and every unused-record branch.*

Proof. The transcript fixes J , the full latent vector, all arm indices, all nonhidden records, the set H , its ordered positions, and R . In particular, the overflow predicate $\mathbf{1}\{R > n\}$ is the same for s and s' because it contains no data-dependent quantity.

If $R > n$, Proposition A.24 does not call A and returns (12) on both inputs. Hence the two conditional output laws are identical.

Suppose $R \leq n$. If $s = s'$, equation (13) gives identical learner inputs. Otherwise, because replacement adjacency means "differ in at most one ordered record," there is a unique $\ell \in [n]$ with

$$z_a = z'_a \quad (a \neq \ell), \quad z_\ell \neq z'_\ell. \quad (16)$$

At every nonhidden position, the two constructed inputs use the same fixed record $T_{I_r}(W_r)$. At hidden position h_a , they use $T_J(z_a)$ and $T_J(z'_a)$. Therefore:

If $\ell > R$, no hidden occurrence uses the changed external record, so all m positions coincide.

If $\ell \leq R$, every position except h_ℓ coincides, while position h_ℓ contains $T_J(z_\ell)$ versus $T_J(z'_\ell)$. Because ϕ_J is injective, T_J is injective on labeled records, so these two records are distinct. The learner inputs differ in exactly one ordered position.

Every external record is thus used at most once. When $R = 0$, every real record is unused and the learner inputs are equal. When $R = n$, all n real records are used exactly once, so any genuine

one-record change affects exactly one learner-input position. Intermediate R uses precisely the prefix $z_1, \dots, z_R$, and changes in $z_{R+1}, \dots, z_n$ have zero effect. This proves the claimed exhaustive dichotomy. $\square$

Proposition A.25 (Exact inherited privacy after restriction and transcript mixing). *Under Assumption 2, Proposition A.24, and Lemma A.14, the kernel B_Π is replacement-adjacency $(\varepsilon_0, \delta_m)$ -DP on ordered n -samples. Both directions of (T2) hold for every event in the full output cube. The pair is inherited without composition, group privacy, amplification, or any additive overflow term; the proof remains valid at $\varepsilon_0 = 1$ and $\delta_m = 0$.*

Proof. Fix replacement-adjacent $s, s' \in \mathbb{Z}_N^n$, an event $G \subseteq \mathbb{V}_N$, and an internal transcript ω . Let $K_\omega(G|s)$ be the conditional output probability of the construction with only the original randomness of A left unfixed.

On $R > n$, both conditional outputs equal g_0 , so

$$K_\omega(G | s) = K_\omega(G | s') = \mathbf{1}\{g_0 \in G\}. \quad (17)$$

On $R \leq n$, Proposition A.24 and (7) give

$$K_\omega(G | s) = A(\rho_J^{-1}(G) | F_\omega(s)), \quad K_\omega(G | s') = A(\rho_J^{-1}(G) | F_\omega(s')). \quad (18)$$

The event in (18) is measurable, and Lemma A.14 says that the two inputs to A are equal or replacement-adjacent. Applying both primitive inequalities (C1), and observing that (17) satisfies the same inequalities because $\varepsilon_0 \geq 0$ and $\delta_m \geq 0$, yields transcript by transcript

$$K_\omega(G | s) \leq e^{\varepsilon_0} K_\omega(G | s') + \delta_m, \quad (19a)$$

$$K_\omega(G | s') \leq e^{\varepsilon_0} K_\omega(G | s) + \delta_m. \quad (19b)$$

Let ν be the law of the input-independent kernel transcript (14) under the internal implementation (2), not the outer analytical completion from Lemma A.13. By construction, the same probability law ν is used for every real input. Averaging (19a) gives

$$\begin{aligned} B_\Pi(G | s) &= \int K_\omega(G | s) \nu(d\omega) \\ &\leq e^{\varepsilon_0} \int K_\omega(G | s') \nu(d\omega) + \int \delta_m \nu(d\omega) \\ &= e^{\varepsilon_0} B_\Pi(G | s') + \delta_m. \end{aligned} \quad (20)$$

Averaging (19b) proves the reverse direction. The additive term remains δ_m , rather than being multiplied by the number of transcripts or positions, because ν has total mass one. Only one invocation of the one-record privacy inequality occurs for each transcript; the changed real record never appears twice. The overflow branch contributes equality, not a failure event or privacy residual.

No inequality in this derivation requires $\varepsilon_0 < 1$, division by δ_m , or a positive additive defect. Hence $\varepsilon_0 = 1$ is included verbatim, and at $\delta_m = 0$ equations (19a)–(20) are the corresponding pure-DP inequalities. This proves (T2) at the exact inherited parameters. $\square$

Proposition A.26 (Exact one-arm baseline specialization). *Under Assumptions 1, 2, and 3, the prior (T1), and Propositions A.24 and A.25, if $k = 1$, then the construction satisfies*

$$J = 1, \quad I_r = 1 \ (r \in [m]), \quad R = m < n = 2m + 12. \quad (21)$$

It never overflows, performs no nonhidden simulation, sends the first m external records once each and in order to A , ignores the remaining $n - m$ records, and remains exactly $(\varepsilon_0, \delta_m)$ -DP. The latent vector is the singleton iid draw from Π .

Proof. When $k = 1$, both the uniform hidden arm and every uniform arm index equal one deterministically. Thus $H = [m]$, $h_a = a$, and $R = m$. Since m is an integer,

$$n = \lceil 2m \rceil + 12 = 2m + 12 > m, \quad (22)$$

so the overflow branch is impossible. There is no arm $j \neq J$, hence no draw in (11). Equation (13) reduces exactly to

$$F_\omega(s) = (T_1(z_1), \dots, T_1(z_m)). \quad (23)$$

The records $z_{m+1}, \dots, z_n$ are unused. A replacement among the first m records changes exactly one position in (23); a replacement among the last $n - m$ records changes none. Proposition A.25 therefore applies without an overflow or arm-simulation branch and retains the primitive pair exactly.

Lemma A.13 gives $\mathbf{E} = (E_1) \sim \Pi$; since $J = 1$ is constant, the required independence is automatic. Thus the construction reduces to the ordinary one-chain restriction interface, not to a stopped, conditional, proper, or remainder-bearing surrogate. This is the exact baseline mechanism consumed by the later $k = 1$ theorem specialization. □

Proposition A.27 (Hidden-arm kernel with exact inherited privacy). *Under Assumptions 1–3 and the prior in (T1), the construction below defines a measurable kernel*

$$B_\Pi : ([N] \times \{0, 1\})^n \rightsquigarrow \{0, 1\}^{[N]}$$

and an iid latent vector $\mathbf{E} \sim \Pi^k$ independent of the uniform hidden arm J . For every replacement-adjacent s, s' and output event G ,

$$B_\Pi(G \mid s) \leq e^{\varepsilon_0} B_\Pi(G \mid s') + \delta_m, \quad B_\Pi(G \mid s') \leq e^{\varepsilon_0} B_\Pi(G \mid s) + \delta_m.$$

The pair is inherited without composition, group privacy, amplification, or an overflow residual. At $k = 1$, $R = m < n = 2m + 12$, there is no overflow or nonhidden-arm simulation and the same conclusions hold exactly.

Proof. Fix the prior (T1). Lemma A.13 samples the latent experiments from the hard prior with exact product law Π^k independent of the uniform hidden arm. It also proves the outer prior-average completion in which the real experiment occupies coordinate J ; because that coordinate is unused by the kernel, this completion leaves the kernel law unchanged and supplies the iid common-experiment vector required for the subsequent coupling.

Proposition A.24 then defines B_Π on every ordered labeled input of size n . It uses only data-independent internal coins and fresh nonhidden records. Its output is the full binary restriction $\rho_J(h)$, whose measurability follows from finitely many coordinate evaluations. Consequently, arbitrary improper and computationally unrestricted outputs of A remain legal, and no threshold projection or properness assumption enters.

Lemma A.14 proves the exact transcript map. Overflow is decided without the input and returns a common constant. Off overflow, the prefix records $z_1, \dots, z_R$ are injected once each into the ordered hidden occurrences. A changed unused record causes equal learner inputs, while a changed used record causes exactly one replacement. The proof explicitly includes $R = 0$, $R = n$, $R > n$, and every intermediate unused-record branch.

Proposition A.25 applies the primitive two-sided DP inequalities to the measurable restricted-output event for each fixed transcript, then averages over the same input-independent transcript law. The additive defect integrates once to δ_m ; it is not multiplied by m , R , k , or the number of transcripts. This proves exactly $(\varepsilon_0, \delta_m)$ -DP, including $\varepsilon_0 = 1$ and $\delta_m = 0$, with no group-privacy, composition, amplification, utility, or overflow loss.

Finally, Proposition A.26 proves that $k = 1$ has $R = m < n$, no overflow, no nonhidden simulation, a singleton iid latent vector, and the ordinary one-chain input map with exact inherited privacy. The conclusion of Proposition A.27 is therefore proved, and B_Π is an admissible arbitrary-output competitor for (HP) whenever the prior is available. $\square$

A.7 Grand-pool coupling and overflow

Retain the objects of Proposition A.27:

$$\mathbf{E} = (E_1, \dots, E_k) \sim \Pi^k, \quad J \sim \text{Unif}([k]), \quad \mathbf{E} \text{ independent of } J, \quad (\text{T7.1})$$

and

$$I_1, \dots, I_m \stackrel{\text{iid}}{\sim} \text{Unif}([k]), \quad R = \sum_{r=1}^m \mathbf{1}\{I_r = J\}. \quad (\text{T7.2})$$

On the common probability space constructed below, conditionally on $\mathbf{E} = \mathbf{e} = (e_1, \dots, e_k)$, the ideal sample satisfies

$$\bar{S} \sim \bar{Q}_{\mathbf{e}}^m, \quad \bar{Q}_{\mathbf{e}} = \frac{1}{k} \sum_{j=1}^k (T_j)_{\#} Q_{e_j}, \quad T_j(q, y) = (\phi_j(q), y). \quad (\text{T7.3})$$

The sole coupling defect is the overflow event

$$\mathbf{O} := \{R > n\}. \quad (\text{T7.4})$$

Lemma A.15 (Bernstein bound for the hidden occupancy). *If independent Bernoulli variables have sum R , mean μ , and total variance v , then for $t > 0$,*

$$\Pr\{R - \mu \geq t\} \leq \exp\left(-\frac{t^2}{2(v + t/3)}\right). \quad (\text{C7.1})$$

Proof. This is Bernstein's upper-tail inequality for bounded independent summands; see, for example, Boucheron et al. (2013). The application below has success probability $1/k$, variance at most $\mu = m/k$, and $t = \mu + 12$. $\square$

Lemma A.16 (Next-unused grand pools give the exact iid mixture). *Under the iid latent-vector conclusion of Lemma A.13, there is a probability-space extension on which, conditional on $\mathbf{E} = \mathbf{e} = (e_1, \dots, e_k)$, one draws mutually independent infinite pools*

$$(W_{j,s})_{s \geq 1} \stackrel{\text{iid}}{\sim} Q_{e_j}, \quad j \in [k], \quad (1)$$

by a conditional law independent of J , and independently of the iid uniform arm indices $I_1, \dots, I_m$. Thus the pools are independent of J conditional on $\mathbf{E}$. For every $(a, \mathbf{e}) \in [k] \times \mathcal{E}_{n,N}^k$, on the corresponding conditional fiber (with an arbitrary version on a null fiber), define

$$C_j(r) := \sum_{u=1}^r \mathbf{1}\{I_u = j\} \quad (2)$$

and the r -th ideal learner record is

$$\bar{Z}_r := T_{I_r} \left(W_{I_r, C_{I_r}(r)} \right), \quad r \in [m], \quad (3)$$

then

$$\bar{S} := (\bar{Z}_1, \dots, \bar{Z}_m) \sim \bar{Q}_{\mathbf{e}}^m, \quad \bar{Q}_{\mathbf{e}} = \frac{1}{k} \sum_{j=1}^k (T_j)_{\#} Q_{e_j}. \quad (4)$$

The assertion is an exact ordered-product law. Repeated selections of one arm consume different pool indices and do not create dependence or record reuse.

Proof. For completeness, construct the pools directly in the present finite notation. Put $b := 9n$, so $b \geq 9$, and write $e_j = (t_j, U_j)$ with $U_j = (u_{j,1}, \dots, u_{j,b})$. Because k is finite, $[k] \times \mathbb{N}$ is countable; fix a bijection $\nu : [k] \times \mathbb{N} \rightarrow \mathbb{N}$. On the Lebesgue probability space $[0, 1)$, write

$$\omega = \sum_{r=1}^{\infty} D_r(\omega) b^{-r}, \quad D_r(\omega) \in \{0, \dots, b-1\},$$

using the base- b expansion ending in zeros at the countable set of points having two expansions. If distinct digit positions $r_1, \dots, r_d$ and digits $a_1, \dots, a_d$ are prescribed, then, apart from those null endpoints, the corresponding event is a union of b^{R-d} half-open base- b intervals of length b^{-R} , where $R = \max_i r_i$. Its probability is therefore b^{-d} . Hence $(D_r)_{r \geq 1}$ are mutually independent and uniform on $\{0, \dots, b-1\}$.

Define, for $(j, s) \in [k] \times \mathbb{N}$,

$$W_{j,s}(\omega) := \left(u_{j, D_{\nu(j,s)}(\omega)+1}, \tau_{t_j} \left(u_{j, D_{\nu(j,s)}(\omega)+1} \right) \right).$$

Each $W_{j,s}$ is uniform on the b labeled row occurrences of U_j , and hence has law Q_{e_j} ; duplicate row values merely combine equal atoms. Injectivity of ν and the finite-cylinder calculation show that every finite subfamily of the $W_{j,s}$ factorizes, which is exactly mutual independence for this countable family. This proves (1) without an external product citation.

Finally, $\mathcal{E}_{n,N}^k$ is finite, so assigning this construction on each latent fiber $\mathbf{e}$ defines a conditional probability kernel (with any version on a null fiber). Its law depends only on $\mathbf{e}$. Taking its product with the conditional law of J and the finite law of the arm-index vector proves the asserted conditional independence from J and $\mathbf{I}$. This is only an enlargement of the probability space; every run reads at most m cells from all pools combined.

Fix an arm-index realization $\mathbf{i} = (i_1, \dots, i_m)$. The pairs

$$(i_r, C_{i_r}(r)), \quad r \in [m], \quad (5)$$

are all distinct. Indeed, if $r < s$ and $i_r = i_s = j$, then the occurrence count of arm j has increased by at least one, so $C_j(r) < C_j(s)$; if the arms differ, the first coordinates in (5) differ. Thus the pool cells selected in (3) are mutually independent conditional on $\mathbf{I} = \mathbf{i}$, even when two selected cells happen to take the same record value.

Let $A_1, \dots, A_m$ be measurable subsets of Z_X , and put

$$P_j := (T_j)_\# Q_{e_j}. \quad (6)$$

Conditional independence of the distinct cells gives

$$\Pr\{\bar{Z}_r \in A_r \ \forall r \mid J = a, \mathbf{E} = \mathbf{e}, \mathbf{I} = \mathbf{i}\} = \prod_{r=1}^m P_{i_r}(A_r). \quad (7)$$

Averaging over the k^m equally likely index vectors,

$$\begin{aligned} & \Pr\{\bar{Z}_r \in A_r \ \forall r \mid J = a, \mathbf{E} = \mathbf{e}\} \\ &= \frac{1}{k^m} \sum_{(i_1, \dots, i_m) \in [k]^m} \prod_{r=1}^m P_{i_r}(A_r) \\ &= \prod_{r=1}^m \left(\frac{1}{k} \sum_{j=1}^k P_j(A_r) \right) = \prod_{r=1}^m \bar{Q}_{\mathbf{e}}(A_r). \end{aligned} \quad (8)$$

This proves (4), including independence of the ordered coordinates, not merely equality of their marginals. The calculation is valid for every hidden designation a , and its right side does not depend on a . Averaging (8) over the conditional law of J therefore also proves $\bar{S} \mid \{\mathbf{E} = \mathbf{e}\} \sim \bar{Q}_{\mathbf{e}}^m$, which is the resulting conditional-mixture claim. This uses no permutation argument, exchangeability conclusion, or PAC property. $\square$

Proposition A.28 (Exact realizability of the arm mixture). *Under Assumption 1 and Lemma A.16, fix $\mathbf{e} = (e_1, \dots, e_k)$ with $e_j = (t_j, U_j)$, and put $\mathbf{t} = (t_1, \dots, t_k)$. Then $\bar{Q}_{\mathbf{e}}$ in (4) is exactly realizable by $c_{\mathbf{t}} \in C$:*

$$\bar{Q}_{\mathbf{e}}\{(x, y) : y = c_{\mathbf{t}}(x)\} = 1. \quad (9)$$

This remains exact when empirical multisets contain duplicate values.

Proof. By the definition of Q_{e_j} , every one of its support records, counted with its multiset multiplicity, has the form

$$(q, \tau_{t_j}(q)) \quad \text{for some } q \in [N]. \quad (10)$$

After transport by T_j , this record becomes

$$(\phi_j(q), \tau_{t_j}(q)). \quad (11)$$

Assumption 1 supplies the concept $c_{\mathbf{t}}$ and gives the pointwise identity

$$c_{\mathbf{t}}(\phi_j(q)) = \tau_{t_j}(q) \quad \text{for every } j \in [k], \ q \in [N]. \quad (12)$$

Therefore $(T_j)_\# Q_{e_j}$ assigns probability one to labels from $c_{\mathbf{t}}$, for every j . Averaging those k laws preserves that probability-one statement and proves (9). Duplicate multiset entries change masses only; they do not change the pointwise label identity (12). No approximate transfer, majority label, or properization is present. $\square$

Lemma A.17 (Recordwise coupling with one-use pool indices). *Under Proposition A.24 and Lemma A.16, define the external threshold-domain sample by*

$$S^{\text{real}} := (W_{J,1}, \dots, W_{J,n}). \quad (13)$$

Conditional on $(J, \mathbf{E})$, (13) has exactly the required law $Q_{E_J}^n$. Couple every nonhidden fresh draw in B_Π at position r , where $I_r = j \neq J$, to

$$W_{j,C_j(r)}. \quad (14)$$

This preserves the exact conditional law of the B_Π construction. On $\mathcal{O}^c = \{R \leq n\}$, its constructed learner input

$$S^{\text{con}} = (Z_1^{\text{con}}, \dots, Z_m^{\text{con}}) \quad (15)$$

satisfies

$$Z_r^{\text{con}} = \bar{Z}_r \quad \text{for every } r \in [m], \quad S^{\text{con}} = \bar{S}. \quad (16)$$

Every pool index used in a learner input is used exactly once. The cases $R = 0$ and $R = n$ lie in (16), while $R > n$ is the sole overflow branch on which B_Π makes no call to A .

Proof. Conditional on $(J, \mathbf{E}) = (a, \mathbf{e})$, the first n coordinates of pool a are iid Q_{e_a} , so (13) has product law $Q_{e_a}^n$. This is precisely the external-sample law in the outer completion of Lemma A.13.

For any fixed arm-index vector and hidden designation, the cells in (14), as r ranges over nonhidden positions, have distinct (j, s) indices by (5). They are consequently independent with the correct respective laws Q_{e_j} , and they are independent of the hidden pool. Thus replacing the abstract fresh nonhidden records in the construction by (14) does not alter any conditional or unconditional law of B_Π .

Let the hidden positions be

$$\{r : I_r = J\} = \{h_1 < \dots < h_R\}. \quad (17)$$

At its a -th hidden occurrence h_a ,

$$C_J(h_a) = a. \quad (18)$$

If $R \leq n$, the construction uses external record $z_a = W_{J,a}$ at that occurrence. Hence

$$Z_{h_a}^{\text{con}} = T_J(W_{J,a}) = T_{I_{h_a}}(W_{I_{h_a}, C_{I_{h_a}}(h_a)}) = \bar{Z}_{h_a}. \quad (19)$$

At a nonhidden position r , the coupling (14) gives directly

$$Z_r^{\text{con}} = T_{I_r}(W_{I_r, C_{I_r}(r)}) = \bar{Z}_r. \quad (20)$$

Equations (19)–(20) prove the recordwise equality (16).

The injectivity of the occurrence-index map (5) proves the promised one-use property. In particular, the hidden learner records use exactly the pool cells $W_{J,1}, \dots, W_{J,R}$, once each. The unused external cells $W_{J,R+1}, \dots, W_{J,n}$ are not inserted elsewhere. Pool cells are distinct record instances even if their sampled values coincide, so equality of values does not amount to reuse.

The boundary branches are now literal:

If $R = 0$, there is no hidden occurrence, and (20) proves equality at every position using only nonhidden next-unused cells.

If $R = n$, whenever this event is feasible, (19) uses all first n external cells exactly once and (20) handles every nonhidden position. The condition is $R \leq n$, so equality at the boundary is included.

If $R > n$, the ideal sample (3) remains well-defined and continues with new hidden-pool cells beyond index n , but the B_Π procedure calls no learner and returns g_0 . It does not recycle any of the first n records. Thus the single event $\mathcal{O}$, and no per-position event, is the coupling defect.

This proof uses neither the privacy certificate, exchangeability, nor PAC utility. $\square$

Proposition A.29 (Diagonal output coupling and bounded-loss transfer). *Under Proposition A.24 and Lemma A.17, the constructed and ideal calls to the arbitrary randomized Markov kernel A can be coupled so that, on $\mathcal{O}^c$,*

$$H^{\text{con}} = \bar{H} \quad \text{as functions on } X, \quad G^{\text{con}} := \rho_J(H^{\text{con}}) = \bar{G} := \rho_J(\bar{H}) \quad \text{on } [N]. \quad (21)$$

On $\mathcal{O}$, $G^{\text{con}} = g_0$, while the ideal call $\bar{H} \sim A(\bar{S})$ remains defined. Consequently, for every common measurable loss functional ℓ taking values in $[0, 1]$, possibly depending on the shared latent variables,

$$|\mathbb{E}\ell(G^{\text{con}}) - \mathbb{E}\ell(\bar{G})| \leq p_{\text{ov}}. \quad (22)$$

In particular,

$$|\mathbb{E}L_{Q_{E_J}}(G^{\text{con}}) - \mathbb{E}L_{Q_{E_J}}(\bar{G})| \leq p_{\text{ov}}. \quad (23)$$

The same conclusions hold when A is deterministic.

Proof. On $\mathcal{O}^c$, Lemma A.17 gives the same ordered input $S^{\text{con}} = \bar{S}$ to both copies of A . Conditional on that common input and all internal variables, draw one h from the probability law $A(\cdot \mid \bar{S})$, and set both H^{con} and $\bar{H}$ equal to h . This is the diagonal coupling of a probability measure with itself; each coordinate has the required marginal law even for an arbitrary randomized output space. Applying the same restriction ρ_J gives (21).

On $\mathcal{O}$, retain the constructed output $G^{\text{con}} = g_0$ and sample the ideal output from $A(\bar{S})$. No relationship is asserted on this branch. Pointwise on the full coupling space, any common $[0, 1]$ -valued loss therefore satisfies

$$|\ell(G^{\text{con}}) - \ell(\bar{G})| \leq \mathbf{1}_{\mathcal{O}}. \quad (24)$$

Taking expectations proves (22). Zero-one risk lies in $[0, 1]$ for every distribution and every hypothesis, so choosing $\ell(g) = L_{Q_{E_J}}(g)$, with the realized E_J included in the shared context, proves (23). Both one-sided inequalities, including the later-needed

$$\mathbb{E}L_{Q_{E_J}}(G^{\text{con}}) \leq \mathbb{E}L_{Q_{E_J}}(\bar{G}) + p_{\text{ov}}, \quad (25)$$

follow from the absolute bound.

If A is deterministic, its two outputs on the common input are already identical, so the diagonal coupling introduces no additional randomness. No exchangeability statement and no PAC-risk bound has been used. $\square$

Lemma A.18 (Uniform Bernstein overflow estimate). *Under the setting conditions $m, k \geq 1$, the iid uniform arm indices in Proposition A.24, and*

$$n = \left\lceil \frac{2m}{k} \right\rceil + 12, \quad (26)$$

let $R = \sum_{r=1}^m \mathbf{1}\{I_r = J\}$ and $\mu = m/k > 0$. Then $R \sim \text{Bin}(m, 1/k)$, all implications caused by the ceiling in (26) are

$$2\mu + 12 \leq n < 2\mu + 13, \quad \{R > n\} \subseteq \{R - \mu > \mu + 12\} \subseteq \{R - \mu \geq \mu + 12\}, \quad (27)$$

and (OF) holds. The exponent in (OF) has the global minimum $27/2$ at $\mu = 6$ over the entire allowed range $\mu > 0$; there is no omitted small-mean or large-mean regime.

Proof. Conditional on any value of J , the indicators

$$X_r := \mathbf{1}\{I_r = J\}, \quad r \in [m], \quad (28)$$

are independent Bernoulli variables with success probability $1/k$. Their conditional law does not depend on the value of J , so the same binomial law holds unconditionally:

$$R \sim \text{Bin}(m, 1/k), \quad \mathbb{E}R = \mu = \frac{m}{k}, \quad v := \text{Var}(R) = \mu \left(1 - \frac{1}{k}\right) \leq \mu. \quad (29)$$

For every real x , $x \leq \lceil x \rceil < x + 1$. Applying this to $x = 2\mu$ in (26) proves the first pair of inequalities in (27). If $R > n$, then

$$R - \mu > n - \mu \geq (2\mu + 12) - \mu = \mu + 12, \quad (30)$$

which proves both event inclusions in (27). Equivalently, because R, n are integers, $R > n$ means $R \geq n + 1$; the weaker strict inequality (30) is exactly the one needed for the stated constant. In particular, equality $R = n$ is not overflow.

Apply Lemma A.15 with $t = \mu + 12 > 0$. Using (27) and $v \leq \mu$,

$$\begin{aligned} p_{\text{ov}} &= \Pr\{R > n\} \\ &\leq \Pr\{R - \mu \geq \mu + 12\} \\ &\leq \exp\left(-\frac{(\mu + 12)^2}{2(v + (\mu + 12)/3)}\right) \\ &\leq \exp\left(-\frac{(\mu + 12)^2}{2(\mu + (\mu + 12)/3)}\right). \end{aligned} \quad (31)$$

The last direction is correct because replacing v by the larger quantity μ enlarges the positive denominator, decreases the positive exponent magnitude, and hence gives a weaker but still valid upper bound.

It remains to prove the uniform numerical exponent rather than assume it. Define for $\mu > 0$

$$F(\mu) := \frac{(\mu + 12)^2}{2(\mu + (\mu + 12)/3)} = \frac{3(\mu + 12)^2}{8(\mu + 3)}. \quad (32)$$

Direct differentiation gives

$$F'(\mu) = \frac{3}{8} \frac{(\mu + 12)(\mu - 6)}{(\mu + 3)^2}. \quad (33)$$

All factors other than $\mu - 6$ in (33) are positive on $(0, \infty)$. Thus F is strictly decreasing on $(0, 6)$, has its unique stationary point at 6, and is strictly increasing on $(6, \infty)$. Moreover,

$$\lim_{\mu \downarrow 0} F(\mu) = 18, \quad F(6) = \frac{3 \cdot 18^2}{8 \cdot 9} = \frac{27}{2}, \quad \lim_{\mu \rightarrow \infty} F(\mu) = \infty. \quad (34)$$

Hence $F(\mu) \geq 27/2$ globally for every allowed $\mu > 0$. This explicitly covers the fact that m/k can be arbitrarily close to zero when k is large, as well as the unbounded large- μ regime. Substitution into (31) proves

$$p_{\text{ov}} \leq e^{-27/2}. \quad (35)$$

Finally, $\log 2 < 1$ implies $10 \log 2 < 10 < 27/2$, so

$$e^{-27/2} < e^{-10 \log 2} = 2^{-10} = \frac{1}{1024}. \quad (36)$$

Equations (31), (35), and (36) are exactly (OF).

□

Proposition A.30 (One-arm and occupancy boundary certificate). *Under Assumption 1, the Proposition A.27 interface, and Lemma A.16 through Lemma A.18, the following boundary conclusions hold without additional assumptions:*

1. *If $k = 1$, then $J = I_r = 1$, $R = m$, $n = 2m + 12 > m$, $p_{\text{ov}} = 0$, the mixture is the single law $(T_1)_{\#}Q_{E_1}$, and constructed and ideal inputs and outputs agree surely.*
2. *On $R = 0$, no hidden pool cell is consumed; on $R = n$, all first n hidden cells are consumed once; on $R > n$, B_{Π} returns g_0 without calling A , and the ideal sample alone continues to unused pool cells.*
3. *Arbitrarily small positive μ , $\mu = 6$, and arbitrarily large μ are all covered by the same exponent lower bound. Deterministic learner outputs obey the same exact equality and bounded-loss transfer.*

Proof. If $k = 1$, uniformity on the singleton set gives $J = 1$ and $I_r = 1$ at every position, so $R = m$. Since m is an integer,

$$n = \lceil 2m \rceil + 12 = 2m + 12 > m = R. \quad (37)$$

Thus overflow is impossible. Formula (3) gives

$$\bar{S} = (T_1(W_{1,1}), \dots, T_1(W_{1,m})), \quad (38)$$

and the external sample is $(W_{1,1}, \dots, W_{1,n})$. The construction uses its first m records once each, so (38) is also the constructed learner input. There are no nonhidden simulations. Formula (4) reduces to $\bar{Q}_{(e_1)} = (T_1)_{\#}Q_{e_1}$, which is realized by $c_{(t_1)}$ by Proposition A.28. Proposition A.29 then gives sure output equality and zero bounded-loss residual.

The three R -branches are the exhaustive cases proved in Lemma A.17; importantly, $R = n$ belongs to the equality event, while only $R > n$ is charged. The small, minimizing, and large mean regimes follow from the complete sign and limit analysis (33)–(34), not from an assumption that $\mu \geq 6$. The deterministic-output case was established directly in Proposition A.29. These checks preserve the exact one-chain baseline rather than a stopped or remainder-bearing surrogate. $\square$

Proposition A.31 (Ideal realizable mixture and overflow certificate). *Under Assumption 1 and Proposition A.27, there is a coupling with an ideal sample satisfying, conditionally on $\mathbf{E} = \mathbf{e} = (e_1, \dots, e_k)$,*

$$\bar{S} \sim \bar{Q}_{\mathbf{e}}^m, \quad \bar{Q}_{\mathbf{e}} = \frac{1}{k} \sum_{j=1}^k (T_j)_{\#}Q_{e_j},$$

and $\bar{Q}_{\mathbf{e}}$ is realized by the minor concept indexed by the latent thresholds. Constructed and ideal inputs and their coupled outputs agree on $\{R \leq n\}$. With $\mu = m/k$,

$$p_{\text{ov}} := \Pr\{R > n\} \leq \exp\left(-\frac{(\mu + 12)^2}{2(\mu + (\mu + 12)/3)}\right) \leq e^{-27/2} < \frac{1}{1024}. \quad (\text{OF})$$

At $k = 1$ the overflow probability is zero and constructed and ideal runs agree surely.

Proof. Start from the exact iid latent vector and hidden-arm kernel supplied by Proposition A.27. Conditional on any latent vector, Lemma A.16 constructs independent infinite pools and uses the occurrence counter of the selected arm to take the next unused cell. Its product calculation proves that the ideal ordered m -sample is iid from exactly $\bar{Q}_{\mathbf{e}}$, rather than merely having the correct marginals. Proposition A.28 then invokes the minor table pointwise to show that this exact mixture is realized by $c_{(t_1, \dots, t_k)}$.

Lemma A.17 chooses the first n hidden-pool cells as the external real sample and uses the same next-unused cells for every nonhidden simulation. This preserves the law of B_{Π} . The occurrence

identity $C_J(h_a) = a$ proves record-by-record equality of all m learner-input positions when $R \leq n$, with no pool cell or hidden external record reused. It also proves that $R = 0$ and $R = n$ are success branches and that $R > n$ is the only defect branch.

Proposition A.29 diagonally couples the two copies of the arbitrary randomized learner on their common input. Thus their full outputs and their hidden-arm restrictions are equal off overflow. Its pointwise indicator inequality charges at most one on the single overflow event for any loss in $[0, 1]$, yielding the specific zero-one-risk transfer (23) without using PAC utility or exchangeability.

Lemma A.18 proves every ceiling implication, identifies R as binomial, applies the checked Bernstein interface, and minimizes the exact exponent globally. The minimum is $27/2$ at $\mu = 6$, with both arbitrarily small and arbitrarily large means controlled, so (OF) follows uniformly. Proposition A.30 finally certifies that $k = 1$ has zero overflow and exact one-arm equality, and consolidates the $R = 0$, $R = n$, $R > n$, and deterministic-output cases. Together these named results prove exactly Proposition A.31 claim and provide only the ideal mixture, coupling certificate, and (OF). □

A.8 Exchangeability and utility

In the common coupling, let

$$\mathbf{E} = (E_1, \dots, E_k) \sim \Pi^k, \quad J \sim \text{Unif}([k]), \quad \mathbf{E} \text{ independent of } J. \quad (\text{T8.1})$$

For $\mathbf{e} = (e_1, \dots, e_k)$, define

$$\bar{Q}_{\mathbf{e}} = \frac{1}{k} \sum_{j=1}^k (T_j)_{\#} Q_{e_j}, \quad T_j(q, y) = (\phi_j(q), y). \quad (\text{T8.2})$$

Let $\bar{S} \sim \bar{Q}_{\mathbf{E}}^m$ and $\bar{H} \sim A(\bar{S})$, and write

$$\bar{G} := \rho_J(\bar{H}), \quad \rho_j(h)(q) = h(\phi_j(q)). \quad (\text{T8.3})$$

Retain the event and bound

$$\mathbf{O} = \{R > n\}, \quad p_{\text{ov}} = \Pr(\mathbf{O}) \leq e^{-27/2} < \frac{1}{1024}. \quad (\text{T8.4})$$

Define the ideal PAC-good event

$$\mathbf{G} := \{L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) \leq \alpha_0\}. \quad (\text{T8.5})$$

The conditional exchangeability identity proved below is

$$\mathbb{E} \left[L_{Q_{E_J}}(\bar{H} \circ \phi_J) \mid \mathbf{E}, \bar{H} \right] = L_{\bar{Q}_{\mathbf{E}}}(\bar{H}). \quad (\text{EX-c})$$

Assumptions 2 and 3 give, for every realizable Q ,

$$\Pr_{S \sim Q^m, h \sim A(S)} \{L_Q(h) > \alpha_0\} \leq \beta_0. \quad (\text{A8.1})$$

The fixed constants are

$$\alpha_0 = \frac{1}{128}, \quad \beta_0 = \frac{1}{32}. \quad (\text{A8.2})$$

Lemma A.19 (The ideal run is ancillary to the hidden designation). *Under Lemma A.13 and Lemma A.16, if the ideal output is sampled from $\bar{H} \sim A(\bar{S})$, then for every latent vector $\mathbf{e}$ and every $a \in [k]$, the conditional law of $(\bar{S}, \bar{H})$ given $(\mathbf{E}, J) = (\mathbf{e}, a)$ is independent of a . Consequently,*

$$\mathbb{E}[\mathbf{1}\{J = a\} \mid \sigma(\mathbf{E}, \bar{S}, \bar{H})] = \frac{1}{k} \quad \text{almost surely,} \quad (1)$$

and in particular J is conditionally independent of $\bar{H}$ given $\mathbf{E}$. The same remains true after conditioning on any event measurable with respect to $(\mathbf{E}, \bar{H})$ that has positive conditional probability.

Proof. Lemma A.13 gives $\Pr\{J = a \mid \mathbf{E} = \mathbf{e}\} = 1/k$. Lemma A.16 gives, for every a ,

$$\mathcal{L}(\bar{S} \mid \mathbf{E} = \mathbf{e}, J = a) = \bar{Q}_{\mathbf{e}}^m, \quad (2)$$

whose right side contains no a . Conditional on any ideal input s , the output law is the same kernel $A(\cdot \mid s)$, which also contains no hidden-designation argument. Therefore, for every measurable set D of ideal input/output pairs,

$$\begin{aligned} \Pr\{J = a, (\bar{S}, \bar{H}) \in D \mid \mathbf{E} = \mathbf{e}\} \\ = \frac{1}{k} \int \mathbf{1}_D(s, h) A(dh \mid s) \bar{Q}_{\mathbf{e}}^m(ds). \end{aligned} \quad (3)$$

The integral in (3) is the conditional probability of $(\bar{S}, \bar{H}) \in D$ given $\mathbf{E} = \mathbf{e}$, so (3) factorizes. This proves (1) and the asserted conditional independence. If D_0 is measurable with respect to $(\mathbf{E}, \bar{H})$, then intersecting D with D_0 preserves the same factorization. Dividing by a positive conditional probability proves the final conditioning statement.

The argument uses the iid common-experiment construction and the J -free ideal kernel law. A random permutation or an unverified symmetry assertion is not used. $\square$

Proposition A.32 (Exact hidden-arm and mixture-risk identity). *Under Lemma A.19, for every $\mathbf{e} = (e_1, \dots, e_k)$ and every hypothesis $h : X \rightarrow \{0, 1\}$,*

$$\frac{1}{k} \sum_{j=1}^k L_{Q_{e_j}}(h \circ \phi_j) = L_{\bar{Q}_{\mathbf{e}}}(h). \quad (4)$$

Consequently, the ideal random objects satisfy (EX-c) and (EX). This conclusion is exact and has no distribution-shift, arm-uniformity, or surrogate-loss residual.

Proof. For fixed j , the transported arm law is $(T_j)_{\#} Q_{e_j}$. By the definitions of T_j , zero-one risk, and ρ_j ,

$$\begin{aligned} L_{(T_j)_{\#} Q_{e_j}}(h) &= \Pr_{(q, y) \sim Q_{e_j}} \{h(\phi_j(q)) \neq y\} \\ &= L_{Q_{e_j}}(h \circ \phi_j). \end{aligned} \quad (5)$$

Risk is affine in the data distribution. Substituting (T8.2) and then (5) gives

$$L_{\bar{Q}_{\mathbf{e}}}(h) = \frac{1}{k} \sum_{j=1}^k L_{(T_j)_{\#} Q_{e_j}}(h) = \frac{1}{k} \sum_{j=1}^k L_{Q_{e_j}}(h \circ \phi_j), \quad (6)$$

which proves (4) pointwise, including hypotheses whose arm risks are highly nonuniform.

By Lemma A.19, for every j , $\mathbb{E}[\mathbf{1}\{J = j\} \mid \sigma(\mathbf{E}, \bar{H})] = 1/k$ almost surely. Since every arm risk below is measurable with respect to $\sigma(\mathbf{E}, \bar{H})$, linearity of conditional expectation and (4) imply

$$\begin{aligned} \mathbb{E}\left[L_{Q_{E_J}}(\bar{H} \circ \phi_J) \mid \sigma(\mathbf{E}, \bar{H})\right] &= \frac{1}{k} \sum_{j=1}^k L_{Q_{E_j}}(\bar{H} \circ \phi_j) \\ &= L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) \quad \text{almost surely,} \end{aligned} \quad (7)$$

which is (EX-c). Taking total expectations proves (EX). This conditional-expectation formulation does not require a regular conditional distribution at each individual hypothesis value. There is no factor of k , because the identity is the exact uniform mixture average. $\square$

Proposition A.33 (PAC conversion on every ideal-mixture fiber). *Under Assumptions 2 and 1, Lemma A.16, and Proposition A.28, for every latent vector $\mathbf{e}$,*

$$\Pr\{L_{\bar{Q}_{\mathbf{e}}}(\bar{H}) > \alpha_0 \mid \mathbf{E} = \mathbf{e}\} \leq \beta_0, \quad (8)$$

and

$$\mathbb{E}[L_{\bar{Q}_{\mathbf{e}}}(\bar{H}) \mid \mathbf{E} = \mathbf{e}] \leq \alpha_0 + \beta_0. \quad (9)$$

Hence $\Pr(\mathbf{G} \mid \mathbf{E} = \mathbf{e}) \geq 1 - \beta_0 > 0$ and

$$\mathbb{E}L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) \leq \alpha_0 + \beta_0. \quad (10)$$

Proof. Fix $\mathbf{e} = (e_1, \dots, e_k)$, with latent thresholds $(t_1, \dots, t_k)$. Proposition A.28 proves that $\bar{Q}_{\mathbf{e}}$ is realized exactly by $c_{(t_1, \dots, t_k)} \in C$, and Lemma A.16 gives the exact sample law $\bar{S} \sim \bar{Q}_{\mathbf{e}}^m$. Applying the primitive PAC guarantee (A8.1) therefore proves (8).

Let $Z = L_{\bar{Q}_{\mathbf{e}}}(\bar{H}) \in [0, 1]$. Pointwise,

$$Z \leq \alpha_0 \mathbf{1}\{Z \leq \alpha_0\} + \mathbf{1}\{Z > \alpha_0\}. \quad (11)$$

Taking the conditional expectation in (11) and using (8) gives

$$\mathbb{E}[Z \mid \mathbf{E} = \mathbf{e}] \leq \alpha_0 + \Pr\{Z > \alpha_0 \mid \mathbf{E} = \mathbf{e}\} \leq \alpha_0 + \beta_0, \quad (12)$$

which proves (9). Equation (8) also gives the stated lower bound on the good-event probability. Finally, averaging (9) over $\mathbf{E} \sim \Pi^k$ proves (10). The PAC failure is charged once by bounded loss; there is no union bound over arms and no conditioning of the unconditional utility conclusion on $\mathbf{G}$. $\square$

Proposition A.34 (Prior-average risk transfer). *Under Lemma A.13, Proposition A.24, and Proposition A.29, the outer prior-average experiment satisfies*

$$\mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(B_{\Pi}, e)] = \mathbb{E}L_{Q_{E_J}}(G^{\text{con}}) \leq \mathbb{E}L_{Q_{E_J}}(\bar{H} \circ \phi_J) + p_{\text{ov}}. \quad (13)$$

The equality and inequality use the same latent experiment and zero-one-risk target; overflow is the only residual.

Proof. By definition of finite-game risk, if an outer experiment $E_* \sim \Pi$ is followed by $S^{\text{real}} \sim Q_{E_*}^n$ and $g \sim B_\Pi(S^{\text{real}})$, then

$$\mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(B_\Pi, e)] = \mathbb{E}L_{Q_{E_*}}(g). \quad (14)$$

Lemma A.13 realizes this experiment by choosing uniform J , setting $E_J = E_*$, and completing the other coordinates iid from Π . It proves that the resulting vector has law Π^k independent of J , while the external sample still has conditional law $Q_{E_J}^n$. Proposition A.24 and the law-preserving coupling in Proposition A.31 give $g = G^{\text{con}}$ in this representation. Thus the right side of (14) is exactly the middle term of (13).

Proposition A.29 applies to the common measurable loss $L_{Q_{E_J}}(\cdot) \in [0, 1]$ and gives

$$\mathbb{E}L_{Q_{E_J}}(G^{\text{con}}) \leq \mathbb{E}L_{Q_{E_J}}(\bar{G}) + p_{\text{ov}}. \quad (15)$$

By (T8.3), $\bar{G} = \bar{H} \circ \phi_J$, so (15) is exactly (13). The same E_J appears in the constructed risk, the ideal risk, and the finite-game objective; no experiment relabeling or surrogate target is introduced. $\square$

Proposition A.35 (Strict prior-average utility below the hard value). *Under Propositions A.32, A.33, and A.34, Lemma A.18, and Assumption 3, the constructed kernel satisfies (UT) with a strict upper bound below 1/20.*

Proof. Proposition A.34, then (EX) from Proposition A.32, and finally (10) from Proposition A.33 give

$$\begin{aligned} \mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(B_\Pi, e)] &\leq \mathbb{E}L_{Q_{E_J}}(\bar{H} \circ \phi_J) + p_{\text{ov}} \\ &= \mathbb{E}L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) + p_{\text{ov}} \\ &\leq \alpha_0 + \beta_0 + p_{\text{ov}}. \end{aligned} \quad (16)$$

By (A8.2),

$$\alpha_0 + \beta_0 = \frac{1}{128} + \frac{1}{32} = \frac{5}{128}. \quad (17)$$

Lemma A.18 gives $p_{\text{ov}} \leq e^{-27/2} < 1/1024$, and therefore

$$\frac{5}{128} + p_{\text{ov}} < \frac{5}{128} + \frac{1}{1024} = \frac{41}{1024} < \frac{1}{20}, \quad (18)$$

where the final strict inequality is equivalent to $20 \cdot 41 = 820 < 1024$. Equations (16)–(18) prove (UT) exactly. Every defect has been charged once: β_0 for PAC failure and p_{ov} for the single overflow event. Neither term is multiplied by k , m , or the number of arm occurrences. $\square$

Proposition A.36 (Conditional hidden-arm certificate). *Under Lemma A.19, Proposition A.32, Proposition A.33, and $\alpha_0 = 1/128 > 0$, the ideal PAC-good event (T8.5) implies the conditional certificate (AUX). If $k = 1$, its success probability is one, the sole-arm risk is at most α_0 , and Proposition A.30 gives zero overflow and exact ideal/constructed equality.*

Proof. For every latent vector and ideal hypothesis define the nonnegative, $\sigma(\mathbf{E}, \bar{H})$ -measurable arm risks

$$Y_j := L_{Q_{E_j}}(\bar{H} \circ \phi_j), \quad j \in [k]. \quad (19)$$

On the event $\mathbf{G}$, the pointwise identity (4) gives

$$\frac{1}{k} \sum_{j=1}^k Y_j = L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) \leq \alpha_0. \quad (20)$$

Lemma A.19 gives $\mathbb{E}[\mathbf{1}\{J = j\} \mid \sigma(\mathbf{E}, \bar{H})] = 1/k$ almost surely. Conditional Markov, written as its finite-arm sum, therefore yields on $\mathbf{G}$

$$\begin{aligned} & \Pr\{Y_J > 8\alpha_0 \mid \sigma(\mathbf{E}, \bar{H})\} \\ &= \frac{1}{k} \sum_{j=1}^k \mathbf{1}\{Y_j > 8\alpha_0\} \\ &\leq \frac{1}{8\alpha_0 k} \sum_{j=1}^k Y_j \leq \frac{1}{8}. \end{aligned} \quad (21)$$

Proposition A.33 gives $\Pr(\mathbf{G} \mid \mathbf{E} = \mathbf{e}) \geq 1 - \beta_0 > 0$, so the conditioning in (AUX) is legitimate under the canonical fiber law. Because $\mathbf{G}$ is measurable with respect to $\sigma(\mathbf{E}, \bar{H})$, the tower property applied to (21) gives

$$\Pr\{L_{Q_{E_J}}(\bar{H} \circ \phi_J) > 8\alpha_0 \mid \mathbf{E} = \mathbf{e}, \mathbf{G}\} \leq \frac{1}{8}, \quad (22)$$

for every canonical latent fiber, and hence Π^k -almost surely. This is equivalent to (AUX). Assumption 3 gives $8\alpha_0 = 8/128 = 1/16$.

When $k = 1$, identity (4) says that the sole-arm risk equals the mixture risk. Thus on $\mathbf{G}$ it is at most $\alpha_0 < 8\alpha_0$, so (AUX) holds with probability one rather than merely $7/8$. Proposition A.30 additionally gives $p_{\text{ov}} = 0$ and sure equality of the ideal and constructed runs. The auxiliary and utility interfaces therefore reduce to the ordinary one-chain PAC conversion without a stopped, conditional, or remainder-bearing weakening. $\square$

Proposition A.37 (Exchangeability and strict utility). *Under Assumptions 1–3 and Propositions A.27–A.31,*

$$\mathbb{E}L_{Q_{E_J}}(\bar{H} \circ \phi_J) = \mathbb{E}L_{\bar{Q}_{\mathbf{E}}}(\bar{H}), \quad (\text{EX})$$

and

$$\mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(B_{\Pi}, e)] \leq \alpha_0 + \beta_0 + p_{\text{ov}} \leq \frac{5}{128} + e^{-27/2} < \frac{1}{20}. \quad (\text{UT})$$

Conditional on the ideal PAC-good event and each latent vector,

$$\Pr\{L_{Q_{E_J}}(\bar{H} \circ \phi_J) \leq 8\alpha_0 \mid \mathbf{E}, L_{\bar{Q}_{\mathbf{E}}}(\bar{H}) \leq \alpha_0\} \geq \frac{7}{8}, \quad 8\alpha_0 = \frac{1}{16}. \quad (\text{AUX})$$

For $k = 1$ the last success probability is one and overflow is absent.

Proof. Lemma A.13 supplies an iid common latent vector independent of the uniform hidden designation, and Lemma A.16 supplies an ideal learner input whose conditional product law depends on that vector but not on the designation. Lemma A.19 pushes this independence through the arbitrary learner kernel, proving that J is still uniform after the latent vector and ideal output are observed.

Proposition A.32 then proves the exact pointwise equality between the uniform average of transported arm risks and the ideal mixture risk. Combining that deterministic identity with the

conditional uniformity of J gives (EX-c) and (EX) without assuming uniform arm performance or invoking a permutation argument.

Proposition A.28 places every ideal mixture within the primitive distribution-free realizable PAC guarantee. Proposition A.33 applies that guarantee on each latent fiber, charges the unit-bounded risk on the PAC-failure event, and obtains expected mixture risk at most $\alpha_0 + \beta_0$.

Proposition A.34 identifies the prior-average finite-game objective with the constructed hidden-arm risk in the same outer completion and applies the diagonal coupling. The ideal hidden-arm risk is the only main term and p_{ov} is the only residual. Proposition A.35 composes this transfer with (EX) and the PAC expectation bound, then proves $5/128 + e^{-27/2} < 1/20$ by the explicit integer comparison $820 < 1024$. This is precisely (UT).

Finally, Proposition A.36 conditions only on the proved ideal PAC-good event and the iid latent vector. The exact arm average is then at most α_0 , so Markov's inequality over the conditionally uniform designation gives threshold $8\alpha_0 = 1/16$ with probability at least $7/8$. The same proposition proves probability one and zero overflow at $k = 1$. These named results jointly prove the conclusion of Proposition A.37 and provide (EX), (UT), and (AUX) to Proposition A.42 without using the hard-prior lower value or the final contradiction. $\square$

A.9 Rate specialization and contradiction

Let $a_{\text{th}}, a_\delta > 0$ and $N_{\text{th}} \geq 2$ be the constants produced by Proposition A.23. Define

$$C_\Delta = 225(1 + \log 15), \quad c_\delta = \frac{a_\delta}{C_\Delta}, \quad c = \frac{a_{\text{th}}}{4}, \quad (\text{T9.1})$$

and choose an integer $N_0 \geq N_{\text{th}}$ for which

$$\log^* N_0 > \frac{26}{a_{\text{th}}}. \quad (\text{T9.2})$$

For the public parameters put

$$n = \left\lceil \frac{2m}{k} \right\rceil + 12. \quad (\text{T9.3})$$

The hard-prior proposition applies under the four scalar conditions

$$N \geq N_{\text{th}}, \quad n < a_{\text{th}} \log^* N, \quad 0 < \varepsilon \leq 1, \quad 0 \leq \delta \leq \frac{a_\delta}{n^2 \log(en)}. \quad (\text{C9.1})$$

Proposition A.38 (Universal specialization constants and finite-domain threshold). *Under Proposition A.22, let $a_{\text{th}}, a_\delta > 0$ and $N_{\text{th}} \geq 2$ be its universal constants. Define C_Δ, c_δ, c by (T9.1). These constants are positive and universal, and there is an integer $N_0 \geq N_{\text{th}}$ satisfying (T9.2). For every $N \geq N_0$,*

$$N \geq N_{\text{th}}, \quad \log^* N \geq \log^* N_0 > \frac{26}{a_{\text{th}}}, \quad 13 < \frac{a_{\text{th}}}{2} \log^* N. \quad (9.1)$$

The constants are independent of $C, X, k, N, m, \varepsilon_0, \delta_m$, the learner, and every random object.

Proof. Because $\log 15 > 0$,

$$C_\Delta = 225(1 + \log 15) > 0. \quad (9.2)$$

The dependency gives $a_{\text{th}}, a_\delta > 0$, so (T9.1) gives $c_\delta, c > 0$. All three constants depend only on universal constants and displayed numerical factors.

The base-two iterated logarithm is nondecreasing and unbounded on the positive integers. For example, if $T_0 = 2$ and $T_{r+1} = 2^{T_r}$, applying one base-two logarithm to T_{r+1} gives T_r , so $\log^* T_r$ tends to infinity. Choose r with $\log^* T_r > 26/a_{\text{th}}$, and take an integer

$$N_0 \geq \max\{N_{\text{th}}, T_r\}. \quad (9.3)$$

Monotonicity gives the middle inequality in (9.1) for every $N \geq N_0$. Multiplication by $a_{\text{th}}/2 > 0$ gives the last inequality. $\square$

Lemma A.20 (Exact sample and privacy-denominator arithmetic). *Under Assumptions 1 and 2, Proposition A.38, and $N \geq N_0$, define*

$$L := \log^* N, \quad \mu := \frac{m}{k}, \quad n := \lceil 2\mu \rceil + 12. \quad (9.4)$$

Then

$$n \leq 15m, \quad n^2 \log(en) \leq C_\Delta m^2 \log(em). \quad (9.5)$$

If, in addition,

$$m < ckL = \frac{a_{\text{th}}}{4} kL, \quad (9.6)$$

then

$$n < a_{\text{th}} L. \quad (9.7)$$

Proof. Since $m, k \geq 1$, $2m/k \leq 2m$, and $2m$ is an integer. Hence

$$n = \left\lceil \frac{2m}{k} \right\rceil + 12 \leq 2m + 12 \leq 14m \leq 15m, \quad (9.8)$$

where $2m + 12 \leq 14m$ is exactly $12 \leq 12m$. Thus this comparison does not require a large- m qualification.

The natural logarithm is increasing and $m \geq 1$, so $\log(em) = 1 + \log m \geq 1$. From (9.8),

$$\begin{aligned} \log(en) &\leq \log(15em) \\ &= \log(em) + \log 15 \\ &\leq (1 + \log 15) \log(em). \end{aligned} \quad (9.9)$$

The last line uses $\log 15 \leq (\log 15) \log(em)$. Combining $n^2 \leq 225m^2$ with (9.9) gives

$$n^2 \log(en) \leq 225(1 + \log 15)m^2 \log(em) = C_\Delta m^2 \log(em). \quad (9.10)$$

For the strict sample condition, $\text{ceil}(x) < x + 1$ for every real x . Equation (9.6) therefore gives

$$n < 2\mu + 13 < \frac{a_{\text{th}}}{2} L + 13. \quad (9.11)$$

Proposition A.38 gives $13 < (a_{\text{th}}/2)L$, so

$$n < \frac{a_{\text{th}}}{2} L + 13 < a_{\text{th}} L. \quad (9.12)$$

The two strict inequalities come respectively from the strict counterassumption and the strict choice (T9.2). $\square$

Proposition A.39 (Admissible hard prior and unchanged private competitor). *Under Assumptions 1, 2, and 3, Proposition A.38, Lemma A.20, and the temporary counterassumption (9.6), with $N \geq N_0$, the instantiated tuple*

$$(n, N, \varepsilon, \delta) = (n, N, \varepsilon_0, \delta_m)$$

satisfies all four conditions in (C9.1), namely

$$(N \geq N_{\text{th}}), \quad n < a_{\text{th}} \log^* N, \quad 0 < \varepsilon_0 \leq 1, \quad 0 \leq \delta_m \leq \frac{a_\delta}{n^2 \log(en)}. \quad (9.13)$$

Consequently, Proposition A.22 produces

$$\Pi = \Pi_{n, N, \varepsilon_0, \delta_m} \in \Delta(\mathcal{E}_{n, N}) \quad (9.14)$$

satisfying (HP), and Proposition A.24 produces the kernel B_Π , while Proposition A.25 certifies that this same kernel is in the exact $(\varepsilon_0, \delta_m)$ -DP competitor class of the same (HP) inequality.

Proof. Proposition A.38 gives

$$N \geq N_0 \geq N_{\text{th}}. \quad (9.15)$$

Lemma A.20 and (9.6) give

$$n < a_{\text{th}} \log^* N. \quad (9.16)$$

Assumption 3 gives

$$0 < \varepsilon_0 \leq 1, \quad 0 \leq \delta_m \leq \frac{c_\delta}{m^2 \log(em)}. \quad (9.17)$$

Substituting $c_\delta = a_\delta / C_\Delta$ gives

$$\delta_m \leq \frac{a_\delta}{C_\Delta m^2 \log(em)}. \quad (9.18)$$

All denominators are positive because $m, n \geq 1$. Equation (9.10) therefore reverses under reciprocation:

$$\frac{1}{C_\Delta m^2 \log(em)} \leq \frac{1}{n^2 \log(en)}. \quad (9.19)$$

Multiplying by $a_\delta > 0$ and combining with (9.18) yields

$$0 \leq \delta_m \leq \frac{a_\delta}{n^2 \log(en)}. \quad (9.20)$$

Equations (9.15)–(9.17) and (9.20) are exactly (C9.1). This is a comparison of the inherited scalar δ_m , not a definition or renaming of a parameter δ_n . The hard-prior result therefore produces (9.14) at precisely $(n, N, \varepsilon_0, \delta_m)$. With that prior fixed, Proposition A.24 constructs B_Π , and Proposition A.25 proves both DP directions with the same pair $(\varepsilon_0, \delta_m)$. Thus B_Π is a legal competitor in (HP). $\square$

$\square$

Proposition A.40 (Strict hard-prior closure). *Under Assumptions 1, 2, and 3, for every $k \geq 1$, $N \geq N_0$, and learner A in the formalized setting, for the same $(n, N, \varepsilon_0, \delta_m)$, prior Π , and kernel B_Π , Proposition A.39 supplies the instantiated (HP) conditions, Proposition A.22 supplies its $(1/20)$ lower bound, Proposition A.24 produces B_Π , Proposition A.25 certifies exact $(\varepsilon_0, \delta_m)$ -privacy for that kernel, and Proposition A.35 supplies (UT) for the same prior and kernel. Under these named interfaces, if the temporary counterassumption (9.6) is made for the contradiction,*

$$m \geq ck \log^* N = \frac{a_{\text{th}}}{4} k \log^* N. \quad (9.21)$$

Proof. Suppose for contradiction that (9.6) holds. Proposition A.39 supplies the prior (9.14) and the kernel produced by Proposition A.24; Proposition A.25 certifies that kernel as the exact-private competitor. Applying the (HP) conclusion of Proposition A.22 to this competitor gives

$$\frac{1}{20} \leq \inf_{B: (\varepsilon_0, \delta_m)\text{-DP}} \mathbb{E}_{e \sim \Pi} [\mathcal{R}_n(B, e)] \leq \mathbb{E}_{e \sim \Pi} [\mathcal{R}_n(B_\Pi, e)]. \quad (9.22)$$

Proposition A.35, applied to exactly the same prior, kernel, risk, and outer expectation, gives

$$\mathbb{E}_{e \sim \Pi} [\mathcal{R}_n(B_\Pi, e)] \leq \frac{5}{128} + e^{-27/2} < \frac{5}{128} + \frac{1}{1024} = \frac{41}{1024} < \frac{1}{20}. \quad (9.23)$$

The last comparison is equivalent to $820 < 1024$. Equations (9.22) and (9.23) therefore contradict each other. Hence (9.6) is false, which is (9.21). All constants were fixed before quantifying over $C, X, k, N, m, \varepsilon_0, \delta_m$ and A , so the required universal quantifier order is preserved. $\square$

Proposition A.41 (Exact one-chain recovery). *Under Assumptions 1, 2, and 3, Proposition A.39 supplies the instantiated hard-prior conditions and unchanged δ_m , Propositions A.26, A.30, and Proposition A.35 supplies the unconditional (UT) bound, Proposition A.36 supplies only the conditional one-arm certificate, and $N \geq N_0$, setting $k = 1$ specializes Proposition A.40 to*

$$m \geq \frac{a_{\text{th}}}{4} \log^* N. \quad (9.24)$$

Moreover, this is the unrestricted one-chain threshold obstruction on the same finite-experiment, arbitrary-output, zero-one-risk, and exact privacy interfaces: $n = 2m + 12$, $R = m < n$, $p_{\text{ov}} = 0$, and no multi-arm or overflow residual remains.

Proof. At $k = 1$,

$$n = \lceil 2m \rceil + 12 = 2m + 12, \quad J = I_1 = \dots = I_m = 1, \quad R = m < n. \quad (9.25)$$

Proposition A.26 says that there are no nonhidden simulations, the first m real records are used once, the remaining $(m+12)$ external records are unused, arbitrary outputs are restricted only by their one-chain coordinates, and privacy remains exactly $(\varepsilon_0, \delta_m)$. Proposition A.30 says that overflow is impossible, the sole-arm mixture is exactly the original threshold experiment, and ideal and constructed inputs and outputs agree surely. Proposition A.36 says only that the one-arm mixture identity is tautological and the conditional auxiliary success probability is one. The unconditional utility bound used below is supplied by Proposition A.35, while Proposition A.39 supplies the four hard-prior conditions for the unchanged δ_m .

The main contradiction retains this exact specialization. If

$$m < \frac{a_{\text{th}}}{4} \log^* N, \quad (9.26)$$

then, by (9.1) and (9.25),

$$n = 2m + 12 < \frac{a_{\text{th}}}{2} \log^* N + 12 < a_{\text{th}} \log^* N, \quad (9.27)$$

because $12 < 13 < (a_{\text{th}}/2) \log^* N$. The same unchanged δ_m comparison (9.18)–(9.20) applies. Under this temporary counterassumption, Proposition A.39 gives the four instantiated HP conditions explicitly listed in (9.13). Thus (HP) is available for the exact one-chain kernel. Because $p_{\text{ov}} = 0$, Proposition A.35 (rather than the conditional certificate in Proposition A.36) specializes to

$$\mathbb{E}_{e \sim \Pi}[\mathcal{R}_n(B_{\Pi}, e)] \leq \alpha_0 + \beta_0 = \frac{5}{128} < \frac{1}{20}, \quad (9.28)$$

where the last comparison is equivalent to $100 < 128$. This contradicts the same one-chain (HP) value and proves (9.24).

Thus the entry trace, generated object, metric, arbitrary-output scope, privacy pair, and risk target all coincide with the inherited one-chain obstruction. The only losses are the displayed factor $1/4$, the fixed additive twelve records in $n = 2m + 12$, and the explicit denominator constant C_{Δ} ; there is no residual, stopped event, properness restriction, or weaker surrogate. $\square$

Proposition A.42 (Rate specialization and contradiction). *Under Assumptions 1–3, for every $N \geq N_0$ the constants in (T9.1)–(T9.2) give*

$$m \geq \frac{a_{\text{th}}}{4} k \log^* N = ck \log^* N.$$

The same parameter δ_m is used in the hard prior, hidden-arm kernel, and utility bound. At $k = 1$, $n = 2m + 12$, $R = m < n$, overflow is zero, there is no nonhidden simulation, and the exact one-chain privacy and risk interfaces yield $m \geq (a_{\text{th}}/4) \log^ N$.*

Proof. Proposition A.22 provides universal $a_{\text{th}}, a_{\delta}, N_{\text{th}}$ and the exact hard value $1/20$. Proposition A.38 defines

$$C_{\Delta} = 225(1 + \log 15), \quad c_{\delta} = a_{\delta}/C_{\Delta}, \quad c = a_{\text{th}}/4,$$

and chooses $N_0 \geq N_{\text{th}}$ with $\log^* N_0 > 26/a_{\text{th}}$.

Lemma A.20 proves

$$n \leq 15m, \quad n^2 \log(en) \leq C_{\Delta} m^2 \log(em),$$

and, from the negation of the target, $n < a_{\text{th}} \log^* N$. Proposition A.39 combines these inequalities with Assumption 3 to verify every hard-prior condition. Its reciprocal-denominator comparison transfers the same δ_m ; no privacy parameter is renamed or absorbed.

Proposition A.24 produces B_{Π} , and Proposition A.25 certifies it as a competitor in the exact (HP) infimum. Proposition A.40 then invokes Proposition A.35 for the strict (UT) upper value and compares it with the lower value $1/20$ for the same prior-average risk, proving

$$m \geq \frac{a_{\text{th}}}{4} k \log^* N = ck \log^* N.$$

Finally, Proposition A.41, using Proposition A.39, Propositions A.26, A.30, and A.35 for unconditional UT together with A.36 only for the conditional certificate, proves exact $k = 1$ recovery: $n = 2m + 12$, $R = m < n$, zero overflow, exact ideal/constructed equality, exact privacy, and the unrestricted $\Omega(\log^* N)$ conclusion. This completes the proof. $\square$

A.10 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.23 derives the hard prior from the finite empirical obstruction proved in Propositions A.3–A.18. Proposition A.27 then constructs, from that prior and Assumptions 1–3, an arbitrary-output threshold kernel with exactly the inherited privacy pair. Proposition A.31 supplies the realizable ideal-mixture coupling and its sole overflow defect, and Proposition A.37 proves the strict utility inequality (UT) for the same prior, kernel, privacy pair, and risk.

Proposition A.42 verifies, with $C_\Delta = 225(1 + \log 15)$, $c_\delta = a_\delta/C_\Delta$, $c = a_{\text{th}}/4$, and $\log^* N_0 > 26/a_{\text{th}}$, every scalar condition needed to apply the hard prior. Its strict comparison of (HP) and (UT) rules out $m < ck \log^* N$. Hence $m \geq ck \log^* N$, with universal constants independent of all objects listed in Theorem 3.1. $\square$

Proof of Corollary 3.1. The one-arm conclusions in Propositions A.26, A.30, and A.41 give $n = 2m + 12$, $R = m < n$, zero overflow, no nonhidden simulation, exact inherited privacy, and the original one-chain risk interface. Proposition A.42 then gives $m \geq (a_{\text{th}}/4) \log^* N = c \log^* N$. $\square$